CHAPTER 13

Semiclassical Methods in Nuclear Physics

13.1 Introduction

The many methods and formalisms used to deal with the nuclear many-body problem are, as we have seen, powerful, and allow us to explain an enormous number of experimental facts. In actual numerical calculations, where we want to treat realistic situations, the implementation of formalisms such as Hartree–Fock, RPA, or time-dependent Hartree–Fock lead to an enormous amount of work. It is therefore desirable to have qualitatively correct but quick estimates of such quantities as the nuclear ground state energy, density, and giant resonance frequencies.

A method which is able to provide such estimates is very well known from atomic physics, where the self-consistent Thomas–Fermi theory yields surprisingly good results [Go 48]. For a long time in nuclear physics, Thomas–Fermi theory [SB 63, Be 71, MS 69] was not exploited very much because, as we will see, it is not very well suited to handle many-body systems with very short-ranged two body forces in its primitive form. However, recently there have been some further developments of the Thomas–Fermi theory [Ki 67, BR 71b, Gr 72, BB 72, 74, BZ 73, BJC 76, BCK 76, GS 77, CJB 77, DBS 78] which make its application to nuclear physics possible, and it has become known as extended Thomas–Fermi (ETF) theory. It turns out that the corresponding semiclassical expressions represent the different quantities on the average, that is, quantum fluctuations have been averaged out. There exists, therefore, a close analogy to the Strutinsky smoothing. It is well known that if one applies a Strutinsky

smoothing to modern Hartree–Fock calculations [BQ 75a, b], the corresponding liquid drop part agrees very well with the empirical Bethe–Weizsäcker mass formula. It is the intention of the semiclassical theory explained in this chapter to set up a self-consistent procedure for the averaged quantities (density, etc.), which allows us to calculate the liquid drop part contained in Hartree–Fock calculations *without* having to know the full quantum mechanical solution.

Once a semiclassical method is established for static quantities, it is natural to generalize it to the dynamic case [WMW 75, HE 78 and references therein]. In the limit $\hbar \to 0$, the TDHF equation goes over into the collisionless Boltzmann equation. Using an appropriate generalization for the dynamic case of the semiclassical expansion of the density and kinetic energy density, we arrive at a fluid dynamic description of the giant resonances which is equivalent to the sum rule results we presented in Section 8.7. In this respect, it will be important to note that the giant resonances can be described by a fluid dynamical description of *zero* sound rather than by that of ordinary sound [KB 62, No 64a, Th 61b]. However, the fluid dynamic description of arbitrary nuclear collective motion is at present still the subject of intensive studies.

Naturally, many features of this chapter are related to the liquid drop picture of the nucleus (Chap. 1).

13.2 The Static Case

13.2.1 The Thomas–Fermi Theory

The Thomas–Fermi theory [Th 27, Fe 28], together with its extensions, is the semiclassical treatment of nuclear physics in its independent particle or Hartree–Fock approximation. We must therefore consider a single-particle Hamiltonian

$$H = -\frac{\hbar^2}{2m}\Delta + V, \qquad H\varphi_n = \epsilon_n \varphi_n, \tag{13.1}$$

in which the potential V is, in principle, the self-consistent Hartree–Fock potential (5.34). In what follows, however, we shall often assume for the purpose of demonstration that the particles are simply moving independently in a given potential well, like, for example, the Woods–Saxon potential.

Before we go ahead, it is useful to give the definitions of quantities whose semiclassical development we want to study:

(i) The single-particle *propagator*

$$C^t = e^{-iHt/\hbar}, \tag{13.2}$$

which is the solution of the time-dependent Schrödinger equation,

$$\int d^3r''\left[i\hbar\frac{\partial}{\partial t}\delta(\mathbf{r}-\mathbf{r}'') - H(\mathbf{r},\mathbf{r}'')\right] C^t(\mathbf{r}'',\mathbf{r}') = 0 \qquad (13.3)$$

with the initial condition

$$C^{t=0}(\mathbf{r},\mathbf{r}') = \delta(\mathbf{r}-\mathbf{r}'). \qquad (13.4)$$

With (13.1) the propagator can also be written in the form, *

$$C^\beta(\mathbf{r},\mathbf{r}') = \sum_n \varphi_n(\mathbf{r})\varphi_n^*(\mathbf{r}')e^{-\beta\epsilon_n}, \qquad (13.5)$$

where for convenience we set $\beta = it/\hbar$.†

(ii) The trace of the propagator (*partition function*†; the factor of two takes here and henceforth care of the spin):

$$Z(\beta) = \mathrm{Tr}\, C^\beta = 2\int d^3r\, C^\beta(\mathbf{r},\mathbf{r}). \qquad (13.6)$$

(iii) The *spectral density* matrix

$$g^\epsilon(\mathbf{r},\mathbf{r}') = \langle \mathbf{r}|\delta(\epsilon - H)|\mathbf{r}'\rangle = \sum_n \varphi_n(\mathbf{r})\varphi_n^*(\mathbf{r}')\delta(\epsilon-\epsilon_n), \qquad (13.7)$$

in terms of which the propagator (13.2) is given by its Laplace transform

$$C^\beta(\mathbf{r},\mathbf{r}') = \int_0^\infty d\epsilon\, e^{-\beta\epsilon} g^\epsilon(\mathbf{r},\mathbf{r}') \equiv \mathfrak{L}_{\epsilon\to\beta}\left[g^\epsilon(\mathbf{r},\mathbf{r}')\right], \qquad (13.8)$$

where we have assumed that the energy origin is at the bottom of the well. The spectral density matrix is therefore given as the inverse Laplace transform of the propagator (for properties of the Laplace transforms, see [PB 55, AS 65]):

$$\begin{aligned} g^\epsilon(\mathbf{r},\mathbf{r}') &= 2\mathfrak{L}^{-1}_{\beta\to\epsilon}\left[C^\beta(\mathbf{r},\mathbf{r}')\right] \\ &= \frac{2}{2\pi i}\int_{c-i\infty}^{c+i\infty} d\beta\, e^{\epsilon\beta} C^\beta(\mathbf{r},\mathbf{r}') \qquad (c>0). \end{aligned} \qquad (13.9)$$

(iv) The *density of states*

$$g(\epsilon) = \mathrm{Tr}\,\delta(\epsilon - H) = \mathfrak{L}^{-1}_{\beta\to\epsilon} Z(\beta). \qquad (13.10)$$

(v) The *density matrix*

$$\rho(\mathbf{r},\mathbf{r}';\lambda) = \sum_{n=1}^\infty \varphi_n(\mathbf{r})\varphi_n^*(\mathbf{r}')\Theta(\lambda-\epsilon_n) = 2\cdot\mathfrak{L}^{-1}_{\beta\to\lambda}\frac{C^\beta(\mathbf{r},\mathbf{r}')}{\beta}, \qquad (13.11)$$

* In principle, we also should have spin indices s, s'. For simplicity, we treat here only spin saturated systems (cf. Sec. 5.6.2) and will take into account the spin only by a statistical factor of two.

† The choice of β reminds one of an inverse temperature, and indeed if we had $\beta = 1/k_B T$, where k_B is Boltzmann's constant and T the temperature, (13.6) would represent the partition function of statistical mechanics. However, we have to stress the point that here we always deal with a system at $T=0$.

where Θ is the unit step function, λ is the Fermi energy, and we have used a simple property of Laplace transforms (see [AS 65, Chap. 29]). The *density* is simply the local part of the density matrix (we will see in the following that one of the main purposes of the semiclassical approximation will be to express essential physical quantities by functionals of this local part):

$$\rho(\mathbf{r}) = \rho(\mathbf{r}, \mathbf{r}). \tag{13.12}$$

(vi) The *kinetic energy density* (5.85)

$$\tau(\mathbf{r}) = \sum_{i=1}^{3} \left[\frac{\partial^2}{\partial x_i \partial x_i'} \rho(\mathbf{r}, \mathbf{r}') \right]_{\mathbf{r}'=\mathbf{r}}. \tag{13.13}$$

(vii) The *particle number*

$$N = \int_0^{\lambda} d\epsilon \; g(\epsilon) = \mathcal{L}^{-1}_{\beta \to \lambda} \left[\frac{Z(\beta)}{\beta} \right]. \tag{13.14}$$

(viii) The *ground state energy*

$$E = \sum_{n=1}^{A} \epsilon_n = \int_0^{\lambda} d\epsilon \, \epsilon g(\epsilon) = \mu N - \int_0^{\lambda} d\epsilon \int_0^{\epsilon} d\epsilon' \; g(\epsilon')$$

$$= \lambda N - \mathcal{L}^{-1}_{\beta \to \lambda} \left[\frac{Z(\beta)}{\beta^2} \right]. \tag{13.15}$$

In Eq. (13.15) we have again assumed that the particles are in a given potential; in the HF case the ground state energy has to be modified somewhat, as shown in Eq. (5.40).

The *Thomas–Fermi theory* (see, e.g., [Th 61b]) can be explained from quite different points of view. Here we introduce it more or less in the usual way, that is the particles at each point in space feel the potential as if it were locally equal to a constant. We will see that this is also a short time (or a high-temperature; cf. footnote page 529) argument [Gr 72]. In order to actually derive the Thomas–Fermi approximation we shall immediately place our consideration in a somewhat broader context. This will permit us later to go beyond the Thomas–Fermi approximation in a natural way.

To this purpose we develop the potential $V(\mathbf{r})$ entering (13.1) around an arbitrary point $\mathbf{r}_0$ in a Taylor series:

$$V(\mathbf{r}) = V(\mathbf{r}_0) + \sum_{i=1}^{3} \left. \frac{\partial V}{\partial x_i} \right|_{\mathbf{r}=\mathbf{r}_0} (x_i - x_{0i})$$

$$+ \frac{1}{2} \sum_{i,j=1}^{3} \left. \frac{\partial^2 V}{\partial x_i \partial x_j} \right|_{\mathbf{r}=\mathbf{r}_0} (x_i - x_{0i})(x_j - x_{0j}) + \cdots. \tag{13.16}$$

If this development is convergent, the solution of (13.3) with (13.1) and (13.16) will give the exact propagator independent of $\mathbf{r}_0$. If we break the development (13.16) off at the nth step the corresponding propagator will

depend on the point $\mathbf{r}_0$ around which we develop the potential, and the corresponding solution

$$C_{\mathbf{r}_0}^{(n)}(\mathbf{r},\mathbf{r}')$$

will only be good for values of $\mathbf{r},\mathbf{r}'$ close to $\mathbf{r}_0$. The best choice will therefore be to take the limit $\mathbf{r},\mathbf{r}'\to\mathbf{r}_0$. Since the point $\mathbf{r}_0$ is arbitrary we will get a propagator $C_{\mathbf{r}_0}^{(n)}(\mathbf{r}_0,\mathbf{r}_0)$ which is valid everywhere, that is, we can now identify $\mathbf{r}_0$ with our original variable $\mathbf{r}$. This procedure only gives us the local part of the propagator; if we want to keep the nonlocal structure of the propagator, the best we can do is not to let $\mathbf{r}$ and $\mathbf{r}'$ individually go to $\mathbf{r}_0$ but to suppose that we developed the potential locally around the center of mass of $\mathbf{r}$ and $\mathbf{r}'$; that is, we define the appropriate solution of (13.3) with (13.16) to be [DBS 78]

$$C^{(n)}(\mathbf{r},\mathbf{r}')=\lim_{\mathbf{r}_0\to\frac{1}{2}(\mathbf{r}+\mathbf{r}')} C_{\mathbf{r}_0}^{(n)}(\mathbf{r},\mathbf{r}'). \tag{13.17}$$

For $n\to\infty$, (13.17) will eventually converge to the exact result if (13.16) is convergent. For n finite, we get a hierarchy of approximations:

$$C^{(0)}, C^{(1)}, C^{(2)},\ldots,$$

which are given in terms of the potential and its n first derivatives.

As we said at the beginning of this section, the Thomas–Fermi approximation consists in considering the potential locally as a constant. This obviously corresponds to a break-off in the expansion (13.16) at the lowest order:

$$\left\{\frac{\partial}{\partial\beta}-\frac{\hbar^2}{2m}\Delta_r+V(\mathbf{r}_0)\right\}C_{\mathbf{r}_0}^{(0)}(\mathbf{r},\mathbf{r}')=0. \tag{13.18}$$

It can be easily verified that the solution of (13.18) is given by

$$C_{\mathbf{r}_0}^{(0)}(\mathbf{r},\mathbf{r}')=\left(\frac{m}{2\pi\hbar^2}\right)^{3/2}\frac{1}{\beta^{3/2}}e^{-(m/2\hbar^2\beta)(\mathbf{r}-\mathbf{r}')^2-\beta V(\mathbf{r}_0)}, \tag{13.19}$$

which has the right boundary condition (13.4). With (13.19) and (13.11) we obtain the Thomas–Fermi approximation for the density matrix [AS 65, Chap. 29]:

$$\rho^{(0)}(\mathbf{r},\mathbf{r}')\equiv\rho^{\mathrm{TF}}(\mathbf{r},\mathbf{r}')=\rho^{\mathrm{TF}}(\mathbf{q})\frac{3}{k_F(\mathbf{q})s}j_1(k_F(\mathbf{q})s) \tag{13.20}$$

with

$$\mathbf{q}=\tfrac{1}{2}(\mathbf{r}+\mathbf{r}'),\qquad \mathbf{s}=\mathbf{r}-\mathbf{r}', \tag{13.21}$$

and j_1 a spherical Bessel function. The local limit of the TF density matrix is given by

$$\rho^{\mathrm{TF}}(\mathbf{r})=\frac{1}{3\pi^2}k_F^3(\mathbf{r})\Theta(\lambda-V(\mathbf{r})), \tag{13.22}$$

and the local momentum is defined by:

$$k_F(\mathbf{r}) = \left[\frac{2m}{\hbar^2}(\lambda - V(\mathbf{r})) \right]^{1/2}. \tag{13.23}$$

It has to be emphasized that the Fermi energy λ is fixed by the condition

$$N = \int d^3r\, \rho^{\mathrm{TF}}(\mathbf{r}) \tag{13.24}$$

and that the approximate density matrix (13.20) has lost some features inherent to the exact one like $\rho^2 = \rho$:

$$\int d^3r_1 \rho^{\mathrm{TF}}(\mathbf{r}, \mathbf{r}_1)\rho^{\mathrm{TF}}(\mathbf{r}_1, \mathbf{r}') \neq \rho^{\mathrm{TF}}(\mathbf{r}, \mathbf{r}') \tag{13.25}$$

and as a consequence ($|D\rangle$ being a Slater determinant and using Wick's theorem but not $\rho^2 = \rho$):

$$\langle D|\hat{N}|D\rangle^2_{\mathrm{TF}} - \langle D|\hat{N}^2|D\rangle_{\mathrm{TF}} = \left(\mathrm{Tr}(\rho^{\mathrm{TF}})\right)^2 - \mathrm{Tr}\left((\rho^{\mathrm{TF}})^2\right) \neq 0. \tag{13.26}$$

Thus whenever we make a variational calculation with semiclassical densities (see below) we have to fix the number of particles, as in BCS theory (Chap. 6), by adding condition (13.24) with a Lagrange multiplier.

In order to compare the Thomas–Fermi density with an exact density, in Fig. 13.1 we show a graphical representation of both densities for a Woods–Saxon potential. We see that in the interior of the nucleus the density is well represented on the "average" but in the surface region it drops off too rapidly, going to zero with zero slope at the "classical turning

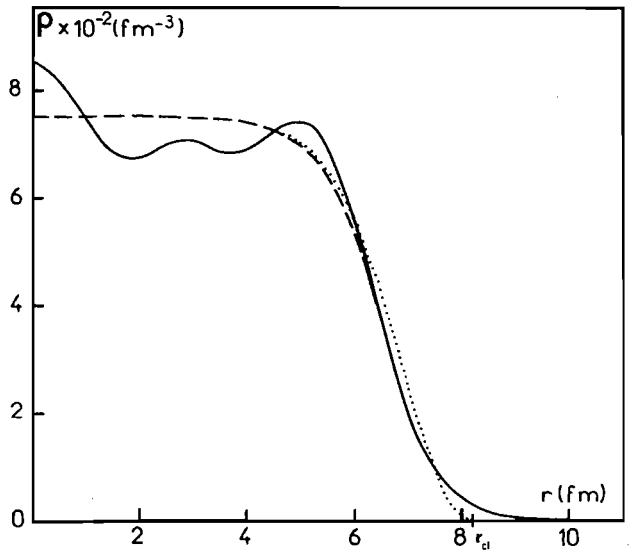


Figure 13.1. Thomas–Fermi approximation (dotted line) to the exact density (full line) in a Woods–Saxon potential for $A = 2N = 184$ particles. The parameters of the well are: $V_0 = -44$ MeV, $a = 0.67$ fm, and $R_0 = 1.27$ fm $A^{1/3}$. The broken line represents the local harmonic approximation of Section 13.2.3. (From [DBS 78].)

point" r_{cl} given by

$$\lambda - V(r_{cl}) = 0. \tag{13.27}$$

Generally speaking, however, the exact single-particle density is surprisingly well represented by such a simple formula like (13.22). This is even more drastically demonstrated if we calculate the ground state energy in Thomas–Fermi approximation using (13.15) with (13.19) and (13.6). Taking the same parameters as those of Fig. 13.1, we obtain from a numerical integration:

$$E_0^{TF} = -3785.8 \text{ MeV}; \qquad E_0^{exact} = -3696.2 \text{ MeV}.$$

The total energy calculated in Thomas–Fermi approximation is only of by $\simeq 3\%$ for heavy nuclei. This means that globally, that is, over the whole space integrated quantities, the Thomas–Fermi approximation is even better than could have been guessed by simply looking at Fig. 13.1.

A comparison of Eqs. (13.11) and (13.9) yields the relation

$$g^{\epsilon}(\mathbf{r}, \mathbf{r}') = \frac{\partial}{\partial \epsilon} \rho(\mathbf{r}, \mathbf{r}'; \epsilon), \tag{13.28}$$

and consequently we obtain for the density of states in Thomas–Fermi approximation:

$$g^{TF}(\epsilon) = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \int d^3r (\epsilon - V(\mathbf{r}))^{1/2} \Theta(\epsilon - V). \tag{13.29}$$

For the harmonic oscillator potential $V = \frac{1}{2} m\omega^2 r^2$, which we will use quite often for demonstration in the following, the integral (13.29) can be evaluated analytically, and we obtain:

$$g_{ho}^{TF}(\epsilon) = \frac{\epsilon^2}{(\hbar\omega)^3}. \tag{13.30}$$

In Fig. 13.2 we compare the integral of this expression, which represents the number of particles with the exact $N(\lambda)$.

We see that again the Thomas–Fermi functions represent roughly the exact ones on the average*; below we will give a more precise discussion on how the average of quantities such as $g(\epsilon)$ shall be defined.

In order to calculate the kinetic energy density (13.13) in Thomas–Fermi approximation we use (13.19) and (13.11) to obtain:

$$\tau^{TF}(\mathbf{r}, \lambda) = \frac{3}{5} \frac{1}{3\pi^2} \left(\frac{2m}{\hbar^2} \right)^{5/2} (\lambda - V)^{5/2} \Theta(\lambda - V) = \frac{1}{5\pi^2} k_F^5 \Theta(\lambda - V). \tag{13.31}$$

Elimination of $\lambda - V$ in (13.31) and (13.22) gives the following relation

* From the observation that $g^{TF}(\epsilon)$ represents the exact density of states in the mean, we can immediately get the usual WKB quantization rule [Ar 65] without using the rather complicated "matching conditions." We do not want to go into more detail of how WKB and TF theory are connected [Vo 77].

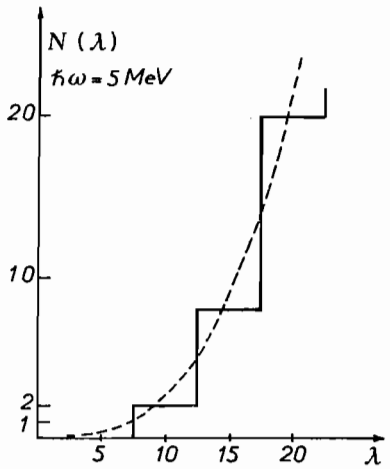


Figure 13.2. Comparison of the exact number of particles as a function of the Fermi energy with the corresponding Thomas–Fermi approximation for the harmonic oscillator.

between the kinetic energy density and the density. [The corresponding relation (5.92) takes into account the isospin.]

$$\tau^{\mathrm{TF}} = \frac{1}{5\pi^2}(3\pi^2 \rho^{\mathrm{TF}})^{5/3}. \tag{13.32}$$

The pure Thomas–Fermi approximation is of little use for the self-consistency problem in nuclear physics because of the short-range two-nucleon interaction, as we stated in the introduction. This can be seen from the fact that for the local part of the Skyrme mean field potential $V[\rho] = a\rho + b\rho^2$ [see Eq. (5.99)], to which the nonlocal parts give only minor changes (a consequence of the short range of the two-body force), the solution for ρ of Eq. (13.22) gives no spatial dependence of ρ whatsoever. For that to happen we will have to extend the pure Thomas–Fermi theory, as will be discussed in the following subsections.

13.2.2 Wigner–Kirkwood $\hbar$ Expansion

13.2.2.1 The Formalism. We now want to show how the Thomas–Fermi results of Section 13.2.1 turn out to be the lowest-order contribution to different quantities, for instance, the single-particle propagator (13.2), in a systematic expansion in powers of $\hbar$. For this purpose it is convenient to go over to the so-called *Wigner transform*. The Wigner transform of a single-particle operator A is defined by [Ar 65]

$$A_{\mathrm{W}} \equiv A(\mathbf{q}, \mathbf{p}) = \int d^3s \, e^{-i\mathbf{p}\mathbf{s}/\hbar} \langle \mathbf{q} + \mathbf{s}/2 | A | \mathbf{q} - \mathbf{s}/2 \rangle, \tag{13.33}$$

with $\mathbf{q}$ and $\mathbf{s}$ given by (13.21). The reason why the Wigner representation is

convenient is due to the fact that, for example, the Hamiltonian (13.1) is equal to its classical counterpart in this representation:

$$H(\mathbf{q},\mathbf{p})=\frac{\mathbf{p}^2}{2m}+V(\mathbf{q}). \tag{13.34}$$

In order to find the completely classical part of the propagator (13.2) we first delvelop its Wigner transform in powers of β:

$$C^{\beta}(\mathbf{q},\mathbf{p})=\left(\sum_{n=0}^{\infty}\frac{1}{n!}(-\beta H)^n\right)_{\mathrm{W}}. \tag{13.35}$$

This means that we have to find the Wigner transform of products of single-particle operators. In Appendix D we derive the following formula:

$$(AB)_{\mathrm{W}}=A(\mathbf{q},\mathbf{p})e^{(i\hbar/2)\overleftrightarrow{\Lambda}}B(\mathbf{q},\mathbf{p}) \tag{13.36}$$

with (the arrows shall indicate in which direction the gradients act):

$$\overleftrightarrow{\Lambda}=\overleftarrow{\nabla}_q\overrightarrow{\nabla}_p-\overleftarrow{\nabla}_p\overrightarrow{\nabla}_q. \tag{13.37}$$

The very useful Eq. (13.36) allows us to easily verify, for example, the well-known result that the classical counterpart of a commutator is proportional to the classical Poisson bracket [Go 59]. With (13.36) we can also calculate successively the terms appearing in the development (13.35). To second order we have, for example:

$$(H^2)_{\mathrm{W}}=H^2(\mathbf{q},\mathbf{p})-\frac{1}{4}\frac{\hbar^2}{2m}\Delta V. \tag{13.38}$$

The $\hbar^2$ correction in (13.38) comes from the fact that the momentum operator does not commute with the potential. It is this $\hbar$ dependence of the single-particle propagator (13.2) which comes from the non-commutativity of the kinetic and potential energy parts of the Hamiltonian in which we are interested and which we intend to expand into a power series in $\hbar$; thus we do not have to worry about the hidden $\hbar^{-1}$ dependence in β (or some other $\hbar$ dependences which may arise in passing from the Wigner to some other representation). It is clear from what we have said that the $\hbar$ *expansion* we are talking about is perfectly manageable with the expansion (13.35) and the product rule (13.36); the powers of $\hbar$ will always be even and equal to the sum of all derivatives in a specific term. The $\hbar$ expansion is therefore simultaneously a *gradient expansion*. The expression to lowest order in $\hbar$ is easily calculated from (13.35), (13.36), and (13.38) to give

$$C^{(0)}(\mathbf{q},\mathbf{p})=e^{-\beta H(\mathbf{q},\mathbf{p})}. \tag{13.39}$$

Calculating the inverse Wigner transform, that is, taking the Fourier transform with respect to $\mathbf{p}$ of (13.39), we find that it is the same as the Thomas–Fermi approximation of the propagator (13.19). It is in this sense that we say that the Thomas–Fermi results of Section 13.2.1 are the expression to lowest order in $\hbar$. At this point it is perhaps useful to reinterpret β as an inverse *temperature* (see footnote page 529) and (13.39)

as the classical statistical operator. Since (13.35) is a simultaneous expansion in $\hbar^2$ and β it is natural that we find for high temperatures—that is, small β—the classical result. It must be emphasized again that in our context β will never play the role of an inverse temperature; on the contrary, we are dealing with a Fermi system at $T=0$. On the other hand, it is important to note that the Thomas–Fermi approximation as well as its extensions, which we will derive in this section, respect the Pauli principle. This can most easily be seen in calculating the density (13.22) from its Wigner transform using (13.39) and (13.11):

$$\rho^{\mathrm{TF}}(\mathbf{q})=\frac{2}{(2\pi\hbar)^3}\int d^3p\, f^{\mathrm{TF}}(\mathbf{q},\mathbf{p})=2\int\frac{d^3p}{(2\pi\hbar)^3}\Theta\left(\lambda-\frac{p^2}{2m}-V(\mathbf{q})\right), \quad (13.40)$$

$$f^{\mathrm{TF}}(\mathbf{q},\mathbf{p})=(\rho^{\mathrm{TF}})_{\mathrm{W}}, \quad (13.41)$$

which is nothing but the sum over all quantum numbers (effective momenta) $p\leqq(2m(\lambda-V))^{1/2}$ from zero up to the Fermi level, each level being occupied twice (spin up, spin down).

The *powers of* β in (13.35) do *not* go along with the *powers of* $\hbar$ and in order to be correct up to $\hbar^2$ we have to develop (13.35) up to β^3. The result is [Wi 32, 34; Ki 33; Ki 57, JB 75]:

$$C^{\beta}(\mathbf{q},\mathbf{p})=\exp\left(-\beta\left(p^2/2m+V(\mathbf{q})\right)\right)\left\{1+\frac{\hbar^2\beta^2}{8m}\left[-\nabla^2V+\frac{\beta}{3}(\nabla V)^2\right.\right.$$

$$\left.\left.+\frac{\beta}{3m}(\mathbf{p}\cdot\nabla)^2V\right]+O(\hbar^4)+\cdots\right\}. \quad (13.42)$$

Expression (13.42) allows us to get semiclassical corrections to all the quantities we considered in Section 13.2.1. With (13.11), we obtain, for example, for the Wigner transform f of the semiclassical density matrix (for a generalization to include a spin orbit term, see [JBB 75]):

$$f_{\mathrm{sc}}(\mathbf{q},\mathbf{p};\lambda)=\Theta\left(\lambda_{\mathrm{sc}}-\frac{p^2}{2m}-V\right)+\frac{\hbar^2}{8m}\left\{-\nabla^2V\,\delta'\left(\lambda_{\mathrm{sc}}-\frac{p^2}{2m}-V(\mathbf{q})\right)\right.$$

$$\left.-\left[\frac{(\nabla V)^2}{3}+\frac{(\mathbf{p}\cdot\nabla)^2V}{3m}\right]\delta''\left(\lambda_{\mathrm{sc}}-\frac{p^2}{2m}-V\right)\right\}+O(\hbar^4), \quad (13.43)$$

where λ_{sc} is determined by (13.14) or (13.24). In (13.43) we ignored the fact that the Laplace inversion of positive powers of β does not exist in the strict sense. In formally creating the different powers of β in (13.42) by differentiation of (13.11) with respect to λ, we obtain the derivatives of the δ function in (13.43). Expression (13.43) can clearly only be used in the sense of a distribution, but we have to study in greater detail in which way the semiclassical expansion (13.42) of the propagator can be useful. From

(13.43) we obtain for the density integrating over **p**:

$$\rho_{sc}(\mathbf{r})=\left\{\frac{1}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}(\lambda_{sc}-V(\mathbf{r}))^{3/2}\right.$$

$$\left.-\frac{1}{24\pi^2}\left(\frac{2m}{\hbar^2}\right)^{1/2}\left[\frac{(\nabla V)^2}{4(\lambda_{sc}-V)^{3/2}}+\frac{\Delta V}{(\lambda_{sc}-V)^{1/2}}\right]+\cdots\right\}\Theta(\lambda_{sc}-V). \tag{13.44}$$

The semiclassical density does not contain any δ functions, but diverges at the classical turning point, and we again have the problem of the meaning of such an expansion.

13.2.2.2 Applicability and Interpretation of the $\hbar$ Expansion. In order to study the validity of the expansion (13.42) and (13.43) it is very useful to investigate some examples for which the propagator is known analytically:

(i) For a one-dimensional *linear potential* $V_1=ax$ it can be checked that the following form of the propagator

$$C^{(1)}(x,x')=\left(\frac{m}{2\pi\hbar^2}\right)^{1/2}\frac{1}{\beta^{1/2}}e^{-(m/2\hbar^2\beta)s^2}e^{-\beta V_1(q)+(\hbar^2/24m)\beta^3[V_1']^2}$$

$$=C^{(0)}(x,x')e^{(\hbar^2/24m)\beta^3V_1'^2} \tag{13.45}$$

fulfills the Schrödinger equation (13.3), where V_1' means the derivative of V_1 and $C^{(0)}(x,x')$ is the one-dimensional analogue to the Thomas–Fermi approximation (13.39) of the single-particle propagator. The Wigner transform of (13.45) affects only $C^{(0)}(x,x')$ and the semiclassical expansion of the remaining factor has to agree with the expression within brackets of (13.42); this can readily be verified. It is also clear that the expansion (13.42)—for this example—is completely valid and it converges in the whole β and $\hbar$ plane. For the calculation of physical quantities [see Eqs. (13.7–13.15)], however, we have to take the inverse Laplace transform of expansions like (13.42) and the problem which was discussed in connection with (13.43) arises. This can be studied in more detail for our example (13.45). Using (13.40) and the folding theorem for Laplace transforms [PB 55, AS 65] in the case of the linear potential we obtain for the density:

$$\rho^{(1)}(q,p;\lambda)=2^{2/3}\sigma\int_{-\infty}^{\lambda}dE\,\Theta\left(\lambda-E-\frac{p^2}{2m}-V_1(q)\right)\mathrm{Ai}(-2^{2/3}\sigma E) \tag{13.46}$$

with

$$\sigma=(2m)^{1/3}\frac{1}{[\hbar V_1']^{2/3}}$$

and the integral representation of the Airy function [AS 65]:

$$\mathrm{Ai}(-z)=\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}d\beta\,e^{\beta z+(1/3)\beta^3}. \tag{13.47}$$

Changing the variable in (13.47) to $y=2^{2/3}\sigma\beta$ we see that $\sigma\mathrm{Ai}$ is proportional to $\delta(E)$ for $\hbar\to 0$ and we recover, together with (13.46), the first terms of (13.43). The higher terms are also obtained in expanding (13.45) up to $\hbar^2$, which can easily be performed again using the variable y in (13.47). From there we can see that all the troubles come from the fact that (13.46) is nonanalytic in $\hbar$ and therefore an $\hbar$ expansion is not really possible [this is analogous to the case that a Fourier transform of e^{-ax^2} is nonanalytic in a, but we can still get a formal power series in a by using δ-functions and derivatives of δ-functions by a term-by-term transformation of the Taylor expansion of e^{-ax^2} (see also Sec. 10.5 on this point)].

(ii) As a second definite example we will treat the three-dimensional *isotropic harmonic oscillator* potential $V_2=\frac{1}{2}m\omega^2r^2$. It can be checked that the corresponding Schrödinger equation (13.3) is fulfilled by the following expression for the propagator (the derivation can be found in [Kr 64]):

$$C^{(2)}(\mathbf{q},\mathbf{s})=\left(\frac{m}{2\pi\hbar^2}\right)^{3/2}\left[\frac{\hbar\omega}{\sinh(\beta\hbar\omega)}\right]^{3/2}\times\exp\left\{-\frac{m\omega}{\hbar}\left[\mathbf{q}^2\tanh\left(\beta\frac{\hbar\omega}{2}\right)+\frac{\mathbf{s}^2}{4}\coth\left(\beta\frac{\hbar\omega}{2}\right)\right]\right\}. \tag{13.48}$$

In Eq. (13.15) we need the trace of this propagator, which can be easily calculated to be:

$$Z^{(2)}(\beta)=\frac{1}{4}\left[\frac{1}{\sinh\left(\beta\frac{\hbar\omega}{2}\right)}\right]^3. \tag{13.49}$$

In order to calculate $E-\lambda N$ we have to evaluate the inverse Laplace transform of $\beta^{-2}Z(\beta)$. This may be done for $Z^{(2)}$ of (13.49) by a contour integration and closing the contour to the left ($\lambda>0$). We thus get contributions from each pole of the partition function $Z^{(2)}$ along the imaginary β axis. We have, therefore, $E-\lambda N$ given by the sum of the residues of the function $\exp(\beta\lambda)Z^{(2)}(\beta)\beta^{-2}$. The pole at $\beta=0$ is of fifth order, whereas the other poles are of only third order. In order to evaluate the residue at the origin, we have to know the Laurent expansion of $Z^{(2)}(\beta)\beta^{-2}$ up to the term $\propto\beta^{-1}$, which is given by:

$$\frac{Z^{(2)}(\beta)}{\beta^2}=\frac{1}{\beta^2}\left[\frac{2}{(\hbar\omega\beta)^3}-\frac{1}{4(\hbar\omega\beta)}+\frac{17}{960}\hbar\omega\beta-\cdots\right]. \tag{13.50}$$

Also calculating the residue to the other poles, we obtain for the energy:

$$E-\lambda N=-\frac{\lambda^4}{12(\hbar\omega)^3}+\frac{\lambda^2}{8\hbar\omega}-\frac{17}{960}\hbar\omega-\sum_{n=1}^{\infty}\frac{2}{(\hbar\omega)^3}(-)^n\left\{\cos\frac{2\pi n\lambda}{\hbar\omega}\left[\left(\frac{\hbar\omega}{2}\right)^2\frac{1}{n^2\pi^2}\left(\left(\frac{\hbar\omega}{2}\right)^2-\lambda^2\right)+6\left(\frac{\hbar\omega}{2}\right)^4\frac{1}{n^4\pi^4}\right]+4\lambda\left(\frac{\hbar\omega}{2}\right)^3\frac{1}{n^3\pi^3}\sin\frac{2\pi n\lambda}{\hbar\omega}\right\}. \tag{13.51}$$

From Eq. (13.51) we see that the contribution of the pole at the origin produces, as a function of λ, a monotonically increasing part, whereas the other poles produce oscillating contributions. These oscillations come from the poles of (13.48) and (13.49) off the real β axis. They are apparently due to the shell structure of the

harmonic oscillator potential since, for the case of the linear potential (13.45), such poles are missing.

Although the harmonic oscillator potential is certainly a very special case, we are tempted to generalize the result to any potential in the sense that the contribution of the singularity of $e^{\beta\lambda}Z(\beta)/\beta^2$ at the origin gives the smooth contribution to the ground state energy, whereas the contribution of other singularities gives the oscillating part.

In spite of the fact that we know very little about the analytical structure of $Z(\beta)$ for an arbitrary potential, these considerations can be substantiated somewhat if we investigate the connection of the Strutinski smoothed energy studied in Sec. 2.9 with the one given just above. For this we consider the *Strutinski smoothed density* of states (2.109):

$$g_{\mathrm{st}}(\epsilon)=\frac{1}{\gamma}\int_0^\infty d\epsilon'\; g(\epsilon')f_M\left(\frac{\epsilon-\epsilon'}{\gamma}\right)$$
$$=\frac{1}{\gamma}\int_0^\infty d\epsilon'\; g(\epsilon')\mathcal{L}^{-1}_{\epsilon-\epsilon'}[f_M(\beta\gamma)]. \tag{13.52}$$

With the definitions (13.6) and (13.8) and the folding theorem for Laplace transforms [PB 55, AS 65] we obtain for the partition function corresponding to (13.52)

$$Z_{\mathrm{st}}(\beta)=Z(\beta)f_M(\beta\gamma), \tag{13.53}$$

where $f_M(\beta\gamma)$ is the Laplace transform of the curvature corrected smoothing factor [(2.111), (2.115)]:

$$f_M(\beta\gamma)=w(\beta\gamma)\sum_{n=0}^{M}\frac{(\beta\gamma)^{2n}}{(2n)!}\left[\frac{\partial^{2n}}{\partial(\beta\gamma)^{2n}}\frac{1}{w(\beta\gamma)}\right]_{\beta=0} \tag{13.54}$$

with

$$w(\beta\gamma)=\frac{1}{\sqrt{\pi}}e^{\beta^2\gamma^2/4}. \tag{13.55}$$

We should recognize that the polynomial in (13.54) represents just the first M terms of the development of $1/w(\beta\gamma)$. In Section 2.9 we showed that the width γ of the Strutinski smoothing must be of the order of $\hbar\omega$ or larger. This causes $f_M(\beta\gamma)$ in (13.54) to be down by five orders of magnitude at the first singularity of (13.49). In other words, Strutinski smoothing simply means cutting out a piece of the partition function (or the propagator) between the first singularities on the positive and negative imaginary β axis, which stays equal to the exact partition function as long as possible and around the poles drops to zero with a width γ. Again we generalize and *assume* that it is true that for any potential with a sequence of bound states to give rise in $Z(\beta)$ to singularities at roughly the mean shell spacing apart. We therefore see that the Strutinski smoothing procedure is essentially equivalent to our definition above of the smooth energy, namely to be that contribution which in (13.15) comes from the singularity of $Z(\beta)$ at the origin, that is, of the behavior of $Z(\beta)$ for small β [Je 73, 76]; this is also confirmed by the numerical examples given below.

The usefulness of the semiclassical expansions such as given by Eq. (13.42) now lies in the fact that they provide simultaneously a Laurent series in $\hbar$ and β of the partition function $Z(\hbar, \beta)$; this is actually all we used to

calculate the contribution to the ground state energy which comes from the singularity of $Z(\beta)$ at $\beta=0$. As a matter of fact, we verify easily that for the harmonic oscillator potential $V(r)=\frac{1}{2}m\omega^2r^2$ we get exactly the expansion (13.50) using (13.42) with $Z=2/(2\pi\hbar)^3\int d^3q\, d^3p\, C(\mathbf{q},\mathbf{p})$ [see Eqs. (13.33) and (13.6)].*

Of course, for an arbitrary potential, the order of the singularity at the origin of $Z(\beta)$ may be very high, forcing us also to push its semiclassical development to high order. There, however, the fact that (13.42) is also an expansion in $\hbar$ helps greatly in the sense that the series is very rapidly convergent. This is clearly demonstrated in Table 13.1, where we show the convergence of the smooth part of the ground state energy for a Woods–Saxon potential. We also give the Strutinski smoothed values and we see the remarkable fact that including fourth order in $\hbar$ gives a convergent value which agrees very well with the Strutinski value. Compared to the exact energies, the contribution from the singularity of $Z(\beta)$ at $\beta=0$ exhausts an enormous amount of the exact total energy, the missing part lying in the parts-per-thousand region. This small amount, however, is not at all negligible, since it contains all the information about the shell structure or quantum oscillations of such quantities as the nuclear density or the total binding energy [see Eq. (13.5)]. For example, it is well known that the smooth part of the total (Strutinski or semiclassical) energy for a continuously deformed nucleus (e.g., quadrupole constraint; see Sec. 7.6) gives only the liquid drop potential curve (Fig. 2.31), whereas quantum oscillations are responsible for such details as double hump barriers, etc., governing the nuclear fission process.

We now have a quick, clear interpretation of our semiclassical or extended Thomas–Fermi method. It allows us to pick up the contribution to the binding energy, which in (13.15) comes from the singularity of $Z(\beta)$ at the origin. If we believe in the generalization of the harmonic oscillator

*It should be noted that the poles of $Z(\beta)$ could be reconstructed from its semiclassical expansion (13.50) in building out of its Laurent series the corresponding Padé approximations, thus also including some of the oscillating contributions to the binding energy. This aspect, however, has not been investigated very much [GS 77].

Table 13.1 Smooth energies in a spherical Woods–Saxon well with the same parameters as those used in Fig. 13.1. All energies are in MeV. The first column gives the nucleon number $A=2N=2Z$. The second, third, and fourth columns give $E_{sc}^{(0)}$, $E_{sc}^{(1)}$, and $E_{sc}^{(2)}$, which correspond to the contributions of the TF term, the $O(\hbar^2)$, and $O(\hbar^4)$ term of Eq. (13.42), respectively. The fifth column gives the total semiclassical energy E_{sc}, while the last column gives the Strutinski energy (from [Je 76]).

A	$E_{sc}^{(0)}$	$E_{sc}^{(1)}$	$E_{sc}^{(2)}$	E_{sc}	E_{st}
164	−3344.7	88.8	1.0	−3254.9	−3256.0 (1.2)
260	−5484.8	119.7	1.1	−5364.0	−5364.4 (0.8)
416	−9049.1	162.5	1.1	−8881.5	−8882.8 (1.2)

result, this contribution corresponds to the smooth part of the energy—and the numerical study for the Woods–Saxon potential supports that—we then have a method which does not need the quantum mechanical result for the calculation of the Strutinski smoothed energy. We should note that it is completely equivalent to use, instead of the semiclassical expansion for $Z(\beta)$ in Eq. (13.15), the ETF result for the density [(13.43) and (13.44)] and calculate $E_{sc}=\mathrm{Tr}(H\rho_{sc})$. It becomes clear from this that the extended Thomas–Fermi method, as it stands, is not suitable to be used as a self-consistent theory; that is, given a $V[\rho]$, Eqs. (13.43) and (13.44) cannot be used to calculate ρ_{sc} self-consistently by iteration because of the divergences at the classical turning point. In the following sections we will see how this drawback of the ETF method can be circumvented.

13.2.2.3 The Functional of the Kinetic Energy Density.* One possibility of disposing of the difficulty that we cannot perform a self-consistent semiclassical calculation is the following: in order to calculate the ground state energy, we need the kinetic energy density $\tau(\mathbf{r})$ of Eq. (13.13). If we could express this kinetic energy density as a functional of the local density, that is, $\tau=\tau[\rho(\mathbf{r})]$, the binding energy would be a functional of $\rho(\mathbf{r})$ alone, since, for example, the Skyrme potential energy (5.87) is given in terms of ρ and τ. A variation of the binding energy with respect to ρ would then give us an equation for $\rho(\mathbf{r})$ alone. Though there exists the theorem by Hohenberg and Kohn [HK 64, MT 77] that the binding energy and thus also the kinetic energy density of a fermion system is a unique functional of $\rho(\mathbf{r})$, it seems to be quite difficult to derive this functional in practice. In our semiclassical framework, however, we are able to construct such a functional which will be valid in the sense that our semiclassical approach is valid, that is, it will ultimately give us the smooth part of (e.g.) the binding energy.

From Eqs. (13.11), (13.13), and (13.42), we are able to construct the semiclassical expansion of τ up to the second order term, which yields [Ki 67, Je 76]:

$$\tau_{sc}(\mathbf{r},\lambda)=\frac{1}{3\pi^2}\left(\frac{2m}{\hbar^2}\right)^{5/2}\left\{\frac{3}{5}(\lambda-V)^{5/2}\right.$$
$$\left.-\frac{1}{8}\frac{\hbar^2}{2m}\left[5\nabla^2V(\lambda-V)^{1/2}-\frac{9}{4}\frac{(\nabla V)^2}{(\lambda-V)^{1/2}}\right]+O(\hbar^4)\right\}\Theta(\lambda-V). \tag{13.56}$$

The expression for the density (13.44) is a function of $\tilde{V}=\lambda-V$, $\nabla\tilde{V}$, and $\Delta\tilde{V}$. Calculating from (13.44) $\nabla\rho$ and $\Delta\rho$ and consistently neglecting higher than second derivatives of the potential, we have three equations: $\rho=\rho(\tilde{V},$

* In this section, we follow to some extent the presentation given in [Br 77].

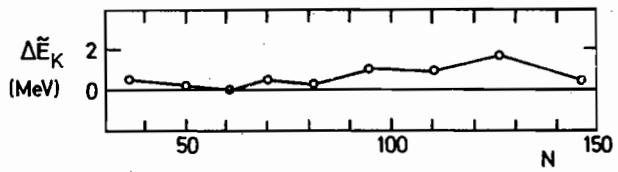


Figure 13.3. The difference $\Delta\tilde{E}_K = \tilde{E}_K - \overline{T}_K$, where $\tilde{E}_K$ is the Strutinsky smoothed kinetic energy and $\overline{T}_K$ is calculated as explained in the text. The values are for a spherical Woods–Saxon potential as a function of neutron number N. (From [Br 77].)

$\nabla\tilde{V}, \Delta\tilde{V}$), $\nabla\rho = \nabla\rho(\cdots)$, and $\Delta\rho = \Delta\rho(\cdots)$, which can be solved for $\tilde{V}$, $\nabla\tilde{V}$, and $\Delta\tilde{V}$; inserting this into (13.56) yields [Ki 67]*:

$$\tau_{sc}[\rho_{sc}] = \frac{3}{5}(3\pi^2)^{2/3}\rho_{sc}^{5/3} + \frac{(\nabla\rho_{sc})^2}{36\rho_{sc}} + \frac{\Delta\rho}{3}$$

$$= \tau^{TF} + \tau^{(2)}. \tag{13.57}$$

[To verify Eq. (13.57) it is easier to calculate the three terms in (13.57) from (13.44) and compare the corresponding result with (13.56).] The first term on the r.h.s. of (13.57) is, of course, the Thomas–Fermi term, which we have already discussed in Section 13.2.1 and is the only one surviving for infinite matter. The following terms take account of the inhomogeneity of the system. The second term is called the Weizsäcker term. The last term in Eq. (13.57) is the divergence of a vector field which vanishes at infinity; from Gauss' theorem this term will not contribute to the kinetic energy, which is the integral over $\tau(\mathbf{r})$. The functional (13.57) is consistent with the semiclassical expansion of ρ [Eq. (13.44)] and τ [Eq. (13.56)] up to order $\hbar^2$. The whole scheme has also been carried out up to order $\hbar^4$ [Je 76, BJ 76b, GV 79]. We do not want to give the quite lengthy expressions here.

According to the derivation, the functional (13.57) is strictly only valid for classically allowed values of $\mathbf{r}$, but it is tempting to assume that the analytical continuation of (13.57) to the region beyond the classical turning point is the correct one for averaged quantities, and in any case we can use (13.57) as a variational ansatz. As a matter of fact, if we put into the functional $\tau_{sc}[\rho_{sc}]$, which is consistent up to order $\hbar^4$, instead of ρ_{sc}, a Strutinsky smoothed density of a spherical Woods–Saxon potential, we see from Fig. 13.3 that the corresponding kinetic energy $\overline{T}_K$ differs only very little from the Strutinsky smoothed one, clearly showing that it is sensible to continue (13.57) to all values of $\mathbf{r}$ [BJC 76].

13.2.2.4 Variational Calculations. Having thus gained some confidence in the functional (13.57), we can adopt the following procedure for the self-consistent determination of a semiclassical density. Using (13.57) we can calculate the smooth

*Sometimes, instead of 1/36, different factors are used in order to correctly reproduce surface energies [Ba 72b].

part of the binding energy $\tilde{E}[\rho_{sc}]=\int d^3r\,\tau_{sc}+\int d^3r v[\rho_{sc}]$ and vary, with respect to ρ_{sc},*

$$\frac{\delta}{\delta\rho_{sc}}\left\{\tilde{E}[\rho_{sc}]-\lambda\int d^3r\,\rho_{sc}(\mathbf{r})\right\}=0, \tag{13.58}$$

where we assure the normalization condition (13.24) with a Lagrange multiplier, as discussed in Section 13.2.1. We should, of course, generalize the theory for Hamiltonians like the one derived from the Skyrme force (5.87) with effective mass and spin orbit terms. This will obviously modify the functional $\tau[\rho]$, which is given in [BJC 76, GV 79]. In order to demonstrate the principle, however, it will be sufficient to neglect effective mass and spin orbit terms and take from the Skyrme potential (5.99) only the local part: $V[\rho]\simeq\frac{3}{8}t_0\rho+\frac{1}{16}t_3\rho^2$. With (13.57) we therefore get the following expression for the smooth part of the binding energy (in the following we replace ρ_{sc} by ρ):

$$\begin{aligned}\tilde{E}[\rho]&=\int d^3r\,\epsilon[\rho],\\ \epsilon[\rho]&=\epsilon_\infty(\rho)+\frac{\hbar^2}{2m}\frac{(\nabla\rho)^2}{36\rho},\\ \epsilon_\infty(\rho)&=\frac{\hbar^2}{2m}\frac{3}{5}(3\pi^2)^{2/3}\rho^{5/3}+\frac{3}{8}t_0\rho^2+\frac{1}{16}t_3\rho^3,\end{aligned} \tag{13.59}$$

where ϵ_∞ is the energy density for infinite nuclear matter, which at saturation ($\rho=\rho_0$) has the value

$$\frac{1}{\rho_0}\epsilon_\infty(\rho_0)=E_0=-16\text{ MeV}\qquad\text{with }\rho_0=0.14\text{ fm}^{-3}. \tag{13.60}$$

To obtain a relatively simple equation, we may try to parametrize $\epsilon_\infty(\rho)$ around its saturating value for infinite nuclear matter in the following way.

$$\frac{1}{\rho}\epsilon_\infty(\rho)\simeq E_0+\frac{K}{18\rho_0^2}(\rho-\rho_0)^2, \tag{13.61}$$

where K, the nuclear incompressibility (5.95), has roughly the value $K\simeq 200$ MeV. The variational procedure (13.58), together with (13.59) and (13.61), leads to the following differential equation for ρ in units of ρ_0 [$\delta(\nabla\rho)^2=2\nabla\rho\delta(\nabla\rho)$ and partial integration, $\lambda\simeq E_0$]:

$$2\frac{\Delta\rho}{\rho}-\frac{(\nabla\rho)^2}{\rho^2}=\frac{1}{a^2}(3\rho^2-4\rho+1), \tag{13.62}$$

with

$$a=\left(\frac{\hbar^2}{2m}\frac{1}{2K}\right)^{1/2}.$$

Equation (13.62) can be solved analytically in one dimension for half-infinite nuclear matter [Br 77]: $\rho(x)=\rho_0[1+\exp(x/a)]^{-1}$. It is easy to get the asymptotic solution of (13.62) requiring that ρ and its derivatives vanish at infinity (no approximation for ϵ_∞ would actually be necessary for this consideration). We then

*This variational principle, in connection with a general energy functional $E[\rho]$, does not correspond to a Ritz variation as discussed in Section 5.6.1.2.

obtain an equation already discussed by Berg and Wilets [BW 56]:

$$r\to\infty: \qquad 2\frac{\Delta\rho}{\rho}-\left(\frac{\nabla\rho}{\rho}\right)^2=a^{-2}, \tag{13.63}$$

which gives for the spherical case

$$\rho(r)\underset{r\to\infty}{\propto}\frac{1}{r^2}e^{-r/a}. \tag{13.64}$$

We see that qualitatively we get the correct behavior for the density, that is, an exponential decay; quantitatively, however, the decay is much too rapid compared to realistic densities, and a numerical solution of (13.62) shows that the exponentially decaying behavior is assumed only quite far out in the tail of the density. In Fig. 13.4, we show density profiles as well as corresponding kinetic energy densities (13.57) that have been calculated by Bohigas et al. [BCK 76], incorporating in the theory a spin-orbit term and the fact that we have different neutron and proton densities. We see that the densities still have a Thomas–Fermi-like behavior, that is, the decay is too steep compared to the real densities. Practically the same results have been obtained by Holzwarth et al. [EH 77]. It is not very astonishing that the densities in Fig. 13.4 still very much resemble Thomas–Fermi densities, since the function (13.57) has been constructed from the extended Thomas–Fermi theory, which yields densities with too steep a surface. The binding energies calculated with the densities of [BCK 76] reproduce the binding energy of the corresponding Hartree–Fock calculation only within 0.4–0.5 MeV per nucleon, indicating that higher-order terms in the functional $\tau[\rho]$ have been neglected (in this respect, see [CJB 77]).

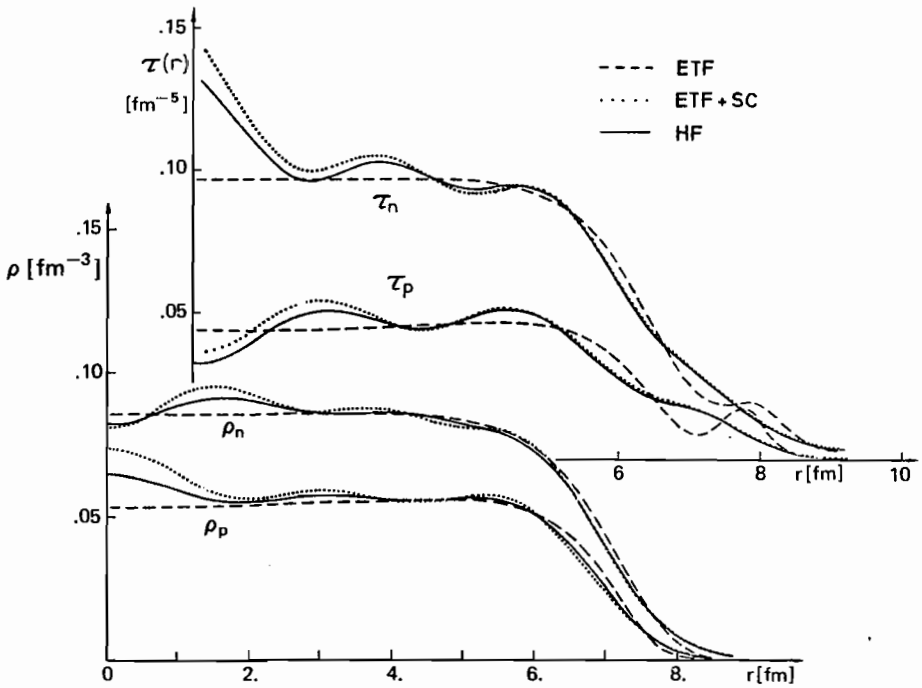


Figure 13.4. Self-consistent semiclassical densities and kinetic energy densities compared to exact results of a Skyrme Hartree–Fock calculation [BCK 76].

In concluding this section, we can say that the semiclassical functional $\tau[\rho]$, together with the Skyrme potential and the variational principle (13.58), establishes a semiclassical theory for the self-consistent calculation of the average nuclear density and the corresponding liquid drop energy. The densities, however, are still too steep in the surface compared to the exact ones, which is a serious drawback if we want to calculate quantities that are sensitive to the tail of the density, for example, the form factors for electron scattering or—in the dynamical generalization—excitation energies of surface oscillations. We will see in the next section how this shortcoming can be remedied.

It is also satisfying that it could be shown (cf. Fig. 13.4) [CJB 77, BCK 76] that starting from these densities (dashed lines) as input to a Hartree–Fock calculation only *one* iteration that is one quantum mechanical calculation (dotted lines) was necessary to reproduce the HF results. It might be of interest that the procedure described in this section is also quite frequently applied in other fields of physics [JM 73, Sec. 8.11.1].

13.2.3 Partial Resummation of the $\hbar$-Expansion

In the last section we saw that the extended Thomas–Fermi approximation is very capable of giving the smooth part of the binding energy. It became clear, however, that the density (13.44) we were implicitly using in this concept is not well defined at or beyond the classical turning point, and that even the density we got out of the density functional formalism has still more or less the Thomas–Fermi-like behavior, that is, compared to the exact density, a surface which is too steep.

In this section, we will show how we can improve on this deficiency. In order to do that, we return to the development (13.16). We had seen that keeping only the first term yields the Thomas–Fermi approximation [(13.19), (13.20)]. It turns out that the propagators $C^{(1)}(\mathbf{r},\mathbf{r}')$, and $C^{(2)}(\mathbf{r},\mathbf{r}')$ can still be given analytically keeping the second and third term, respectively, in (13.16).

The solution keeping only first derivatives in (13.16), which we will call the *linearized case*, can easily be obtained using (13.45) (for simplicity we will only present the solution for the one-dimensional case, but all the results below can straightforwardly be generalized to the three-dimensional case). Using (13.17) and (13.21), we obtain for the propagator [Sw 55, Ba 72b, Gr 72, Bh 77, DBS 78]

$$C^{(1)}(q,s)=\left(\frac{m}{2\pi\hbar^2}\right)^{1/2}\frac{1}{\beta^{1/2}}\exp\left\{-\frac{m}{2\hbar^2\beta}s^2-\beta V(q)+\frac{\hbar^2}{24m}\beta^3\left(\frac{\partial V}{\partial q}\right)^2\right\}. \tag{13.65}$$

It is clear that the approximation (13.65) corresponds to locally replacing the potential by a straight line, and we recover the Thomas–Fermi approximation with $\partial V/\partial q=0$. One can also find the density corresponding to (13.65), which gives an expression analogous to (13.46). We do not want to go into the details of the linearized case, and before we discuss its

relationship to the semiclassical treatment, we want to present the solution of the so-called *locally harmonic case*, where we keep up to second derivatives of the potential in (13.16). Using the result (13.48) as a guide for the harmonic oscillator, we find for the propagator in the one-dimensional case [Gr 72, DBS 78]:

$$C^{(2)}(q,s)=\left(\frac{m}{2\pi\hbar^2}\right)^{1/2}\left[\frac{\hbar\omega(q)}{\sinh(\beta\hbar\omega(q))}\right]^{1/2}$$
$$\times\exp\left\{-\beta\tilde{V}(q)-\frac{m\omega(q)}{\hbar}\left[(\eta(q))^2\tanh\left(\beta\frac{\hbar\omega(q)}{2}\right)+\frac{s^2}{4}\coth\left(\beta\frac{\hbar\omega(q)}{2}\right)\right]\right\}, \tag{13.66}$$

with

$$\omega(q)=\left(\frac{V''(q)}{m}\right)^{1/2},\qquad \eta(q)=\frac{V'(q)}{V''(q)},\qquad \tilde{V}(q)=V(q)-\frac{V'(q)^2}{2V''(q)},$$

where the primes mean derivatives with respect to q. We easily check that for a harmonic oscillator potential (13.66) goes over into the corresponding correct result (13.48). For $V''\to 0$, we also see that (13.16) goes over into (13.65).

In the three-dimensional case, the locally harmonic approximation gets somewhat more complicated, because we have to replace the potential locally by a second-order surface. In order to obtain the solution, we have to find the normal coordinates according to those surfaces in which the Hamiltonian becomes just a sum of three harmonic oscillators. The corresponding propagator is then essentially a product of three propagators of the form (13.66) [DBS 78].

The connection of the linearized (13.65) and the locally harmonic case (13.66) to the extended Thomas–Fermi theory is quite straightforward. In Eq. (13.42) we saw that for the Wigner transform of the propagator, the Wigner–Kirkwood expansion is a simultaneous expansion in powers of $\hbar$ and β for the nonclassical part. It is easy to find the Wigner transforms of (13.65) and (13.66), since they only have a Gaussian dependence in the nonlocality s. Developing, apart from the classical part, the Wigner transform of (13.66) in powers of β, we get back exactly the same series as (13.42), if we put all derivatives higher than the second equal to zero. The same is obviously true for (13.65), where we have to put all derivatives higher than the first equal to zero. The successive approximation scheme [(13.39), (13.65), (13.66)] can therefore also be interpreted in the following way. The TF approximation (13.39) consists of re-summing the series (13.35) for all terms containing no derivatives of the potential; the linearized approximation consists of re-summing all terms containing the potential and its first derivatives, and the locally harmonic approximation in re-summing all the terms containing the potential. its first and second derivatives. It is clear that higher-order derivatives cannot be completely

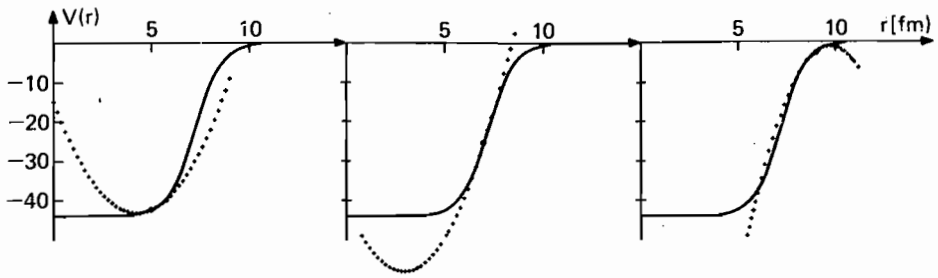


Figure 13.5. Parabolic approximation of the linear Woods–Saxon potential at three different points.

re-summed any more because cubic or higher-order potentials cannot be solved analytically. It is obvious that in terms of powers of $\hbar$ this is only a partial resummation—not all the terms of a given order in $\hbar$ are taken into account. These partial re-summations obviously have desirable features: As discussed above, the expansion (13.42) cannot be Laplace inverted term by term, whereas the expressions (13.65) and (13.66) can be Laplace inverted even analytically [Bh 77, DBS 78]. The partial summation of infinitely many terms can account for important effects, for instance, tunneling into the classically forbidden region, as will be shown below in a numerical example.

Let us study the *geometrical interpretation* of the locally harmonic approximation and study the implications it has for the calculation of the corresponding density. In Fig. 13.5 we show a one-dimensional Woods–Saxon potential and its locally harmonic approximation at three different points. The curves are parabolas with different curvatures in the interior of the inflection point; outside the inflection point the curves are inverted parabolas. In the outside region, we easily imagine that we have penetration into the local barrier, eventually giving rise to quite a good representation of the exponential decay of the density. In the interior the situation is more complicated: The oscillations of the true density (see Fig. 13.1) are global effects, that is, they are due to standing waves in the potential and their wave length is therefore determined by the precise position of the edges of the potential. Since we are only locally representing the potential well, the edges are badly determined and are, as a matter of fact, strongly fluctuating, as can be seen from Fig. 13.5a, b. In the interior, we are therefore introducing spurious shell fluctuations, and all we can hope for is that the density in the interior is well represented on the average, that is, we have to average out the spurious shell fluctuations.

13.2.4 The Saddle Point Method

This average can be done effectively if we perform the necessary Laplace inversions for the density (13.11) and the total energy (13.15) with the help of the saddle point method (SPM) [PB 75, Bh 77]. This consists of

writing

$$G(\lambda)\equiv\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}d\beta\, e^{\beta\lambda}F(\beta)=\frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty}d\beta\, e^{S(\beta)}$$

and expanding $S(\beta)$ around the stationary point β_0 determined by $\partial S/\partial\beta \equiv \dot{S}_0=0$ to second order: $S(\beta)\simeq S_0+\frac{1}{2}\ddot{S}_0(\beta-\beta_0)^2$. Usually this is not sufficient, and we have to introduce corrections to this simple method in expanding S to higher terms in $\beta-\beta_0$. Usually these higher terms are not kept in the exponent but expanded into a Taylor series. Putting together the terms whose order in the derivatives is the same, and including the first correction, we obtain:

$$G(\lambda)=\left(\frac{1}{2\pi\ddot{S}_0}\right)^{1/2}e^{S_0}\left[1+\frac{1}{8\ddot{S}_0^3}\left(\dddot{S}_0\ddot{S}_0-\frac{5}{3}\ddot{S}_0^2\right)+\cdots\right].$$

From this it becomes clear how to construct the second-order correction, which is somewhat tedious but straightforward. However, its inclusion is necessary in most cases in order to gain a convergent result.

For a demonstration of this method we use the exact propagator of the three-dimensional harmonic oscillator (13.48) and calculate with the stationary phase method densities (Fig. 13.6) and total energies. We see that the densities are well-defined average densities in the sense that they pass in a smooth way through the quantum mechanical oscillations. Calculating in the same way the total energy (13.15), we obtain for all deformations the Strutinsky averaged value [BP 73] to within a fraction per thousand, that is, for ground state energies of several thousand MeV the deviation from the Strutinski value is never greater than 1 MeV.

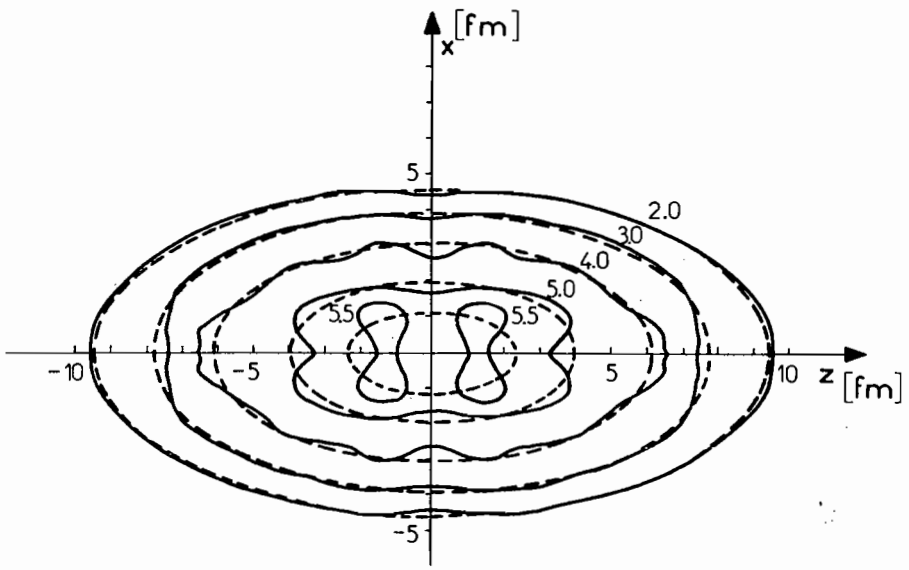


Figure 13.6. Exact densities compared to those obtained by the stationary phase method for a strongly deformed harmonic oscillator. (We are grateful to J. Bartel for the preparation of this figure.)

13.2.5 Application to a Spherical Woods–Saxon Potential

The locally harmonic form (13.66) of the propagator together with the saddle-point method of Section 13.2.4 are the main ingredients of the present theory. In Fig. 13.1 we show a density obtained with the locally harmonic approximation (13.66) for a spherical Woods–Saxon potential by using the stationary phase method for the Laplace inversion. We see the very good agreement of this density with the exact one in the tail region which should be contrasted with the steep surface of the density functional method (Fig. 13.4). Contrary to the ETF theory, the locally harmonic solution allows tunnelling into the classically forbidden region. In the interior this approximation bends over into the TF result, representing there only the average density. The density profile obtained therefore corresponds exactly to the one which we would imagine for a liquid drop. A very stringent test of the validity of the approximation is not only the quality of the density itself, but also, as we have seen, the reproduction of the average of the sum of single-particle energies. With the locally harmonic solution, the average sum of single-particle energies agrees with those of a Strutinski calculation to within 5–10 MeV, depending on the mass number of the nucleus. The theory therefore simultaneously yields very satisfying average energies *and* densities.

The missing 5–10 MeV are most likely due to the neglect of third and fourth derivatives of the potential, since they are known to be important in reproducing the Strutinski value within the Wigner–Kirkwood expansion (see Sec. 13.2.2.2 and Table 13.1). Higher derivatives can be included perturbatively, but we do not want to go into these details here.

The locally harmonic case is also suitable for a self-consistent determination of the *liquid drop parameters* since, taking $V = V[\rho]$ as an input to (13.66), we can set up with (13.11) and (13.13) an iteration procedure for the average density and kinetic energy density. It has been shown by Brack and Quentin [BQ 75a] that Strutinski smoothed Hartree–Fock calculations with the Skyrme force perfectly reproduce the liquid drop parameters of the Bethe Weizsäcker formula. Since the results of the locally harmonic case discussed above are very close to the Strutinski results, it is quite likely that a self-consistent semiclassical treatment using the harmonized theory should give the liquid drop parameters without making the detour over a full Hartree–Fock calculation [BQ 75a]. The feasibility of the method has been shown for a one-dimensional case [EVB 79].

For a complete semiclassical treatment of the self-consistent single-particle problem we must also be able to treat, for instance, the *nonlocal parts* (effective mass) of the Skyrme potential (5.87). This can also be achieved by slightly generalizing the present formalism. However, in order to do this it is preferable to work with Wigner transforms from the beginning and we obtain for the Schrödinger equation (13.3) with (13.36), for example,

$$\left[\frac{\partial}{\partial\beta} + H(\mathbf{q},\mathbf{p})e^{(i\hbar/2)\overleftrightarrow{\Lambda}}\right]C^{\beta}(\mathbf{q},\mathbf{p}) = 0,$$

where $H(\mathbf{q}, \mathbf{p})$ is the Wigner transform of the nonlocal Hamiltonian:

$$H(\mathbf{q}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{q}, \mathbf{p}).$$

The nonlocal Hamiltonian function can be locally approximated in the phase space $\mathbf{q}, \mathbf{p}$ by a second-order surface, analogous to the way this was done for the local case. The corresponding propagator can again be found analytically, and the corresponding densities and energies are of the same quality as in the purely local case [DSB 80].

13.2.6 Semiclassical Treatment of Pairing Properties

Having set up a semiclassical treatment of the self-consistent single-particle density, it seems natural to try a generalization that includes pair correlation (see Chaps. 6 and 7). Since this aspect is far less investigated we will content ourselves here with the solution of the BCS equations (6.54) to lowest order in $\hbar$.

In Chapter 7 we saw that the BCS equation can be written in the form $[\mathcal{K}, \mathcal{R}] = 0$ [Eq. (7.43)]. To lowest order in $\hbar$, the commutator becomes the classical Poisson bracket, as already discussed in the context of Eq. (13.36). Together with the $\hbar \to 0$ limit of $\mathcal{R}^2 = \mathcal{R}$ [Eq. (7.28)], we obtain a set of two algebraic equations for the Wigner transforms of the normal density $\rho(\mathbf{q}, \mathbf{p})$ and the anormal density $\kappa(\mathbf{q}, \mathbf{p})$ [Eq. (7.22)]. The solution leads to the very natural result [cf. (7.61)]

$$\rho(\mathbf{q}, \mathbf{p}) = \frac{1}{2}\left\{1 - \frac{h(\mathbf{q}, \mathbf{p})}{E(\mathbf{q}, \mathbf{p})}\right\},$$

$$\kappa(\mathbf{q}, \mathbf{p}) = \frac{1}{2}\frac{\Delta(\mathbf{q}, \mathbf{p})}{E(\mathbf{q}, \mathbf{p})},$$

$$E(\mathbf{q}, \mathbf{p}) = \sqrt{h^2(\mathbf{q}, \mathbf{p}) + \Delta^2(\mathbf{q}, \mathbf{p})},$$

with $h = \mathbf{p}^2/2m + V(\mathbf{q}) - \lambda$; the Wigner transform of the pair potential $\Delta(\mathbf{q}, \mathbf{p})$ is related to the anormal density by

$$\Delta(\mathbf{q}, \mathbf{p}) = \frac{1}{(2\pi\hbar)^3}\int d^3k\, v(|\mathbf{p} - \mathbf{k}|)\kappa(\mathbf{q}, \mathbf{k}).$$

$v(p)$ is a Fourier transform of the two-body interaction. Thus for the gap equation to lowest order in $\hbar$ we obtain:

$$\Delta(\mathbf{q}, \mathbf{p}) = \frac{1}{2}\frac{1}{(2\pi\hbar)^3}\int d^3k\, v(|\mathbf{p} - \mathbf{k}|)\frac{\Delta(\mathbf{q}, \mathbf{k})}{E(\mathbf{q}, \mathbf{p})}. \tag{13.67}$$

In infinite nuclear matter, where there is no q-dependence, this is an exact equation [FW 71]. By solving Eq. (13.67) independently for each value of $\mathbf{q}$ we treat the nucleus as if locally it were a piece of nuclear matter. This, of course, is the usual Thomas–Fermi assumption [notice the

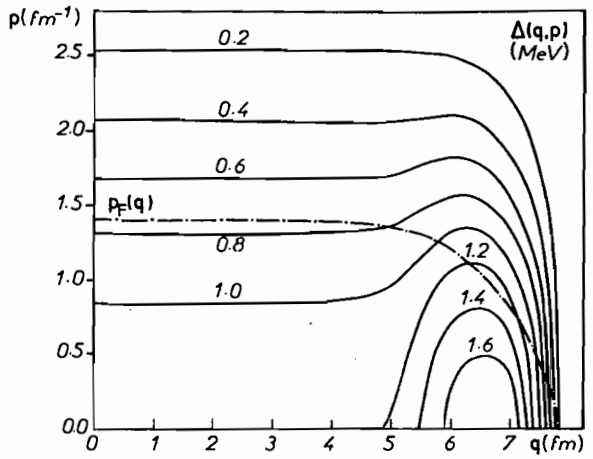


Figure 13.7. The neutron pairing gap in Wigner space, calculated with a Woods–Saxon potential ($V=-49$ MeV, $R_0=6.81$ fm, $a=0.66$ fm) for neutron number 96. The parameters of the two-body force are $V_0=-28$ MeV and $r_0=1.6$ fm.

close analogy to the spirit of the local density approximation for the G-matrix (Sec. 4.3.1)]. For a spherical nucleus, Eq. (13.67) was solved by iteration, taking for the interaction a Gaussian of strength V_0 and range r_0 [BS 80]. In momentum space this reads: $v(p)=-|V_0|(r_0\hbar\sqrt{\pi})^3\exp(-p^2r_0^2/4)$. In Fig. 13.7 we display the solution of $\Delta(q, p)$ for $V_0=-28$ MeV and $r_0=1.6$ fm, and in Fig. 13.8 the solution for κ is shown. We should notice that Δ is peaked around the nucleus surface and that it is exactly zero beyond the classical turning point. The abnormal density as a function of p is very sharply peaked around the local Fermi momentum p_F (13.23). However, the width is largest in the nuclear surface. Since the abnormal energy entering the ground state energy (7.80) is the trace of the product of Δ and κ, the effective values of Δ relevant for the energy are those along the local Fermi momentum $p_F(q)$, indicated by the broken line in Fig. 13.7. In Fig. 13.9, we show a cut of Δ along this line for two values of the force parameters, giving the same pairing energy $E^P\simeq 2$ MeV. From this figure it becomes clear that the pairing phenomenon is essentially a

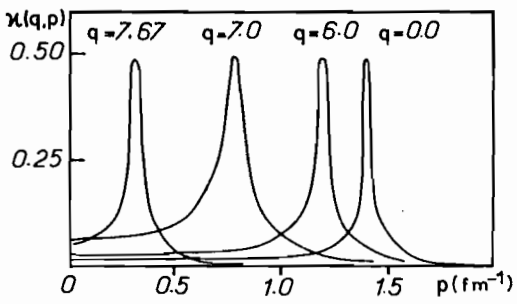


Figure 13.8. The pairing tensor at different positions, q, in the nucleus for the same parameters as in Fig. 13.7.

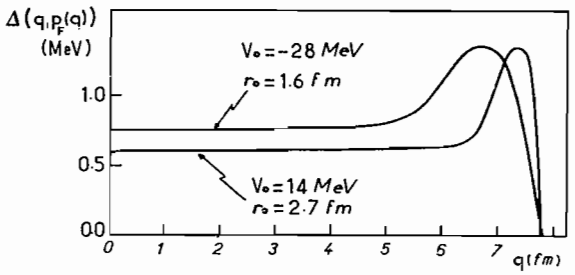


Figure 13.9. The pairing gap along the local Fermi surface [$p_F(q)$ in Fig. 13.7] for two different sets of force parameters.

surface effect for the force used here, a feature which might be important for the transfer of pairs of nucleons.

The interpretation of our results is of course the same as the Thomas–Fermi approach to the normal density: the different quantities are represented on the average, that is, shell corrections in the Strutinsky sense are missing. For example, the exact solution of the pairing gap in Fig. 13.9 should show wiggles superposed to the semiclassical curve, very much like the case for the normal density in Fig. 13.1. With the semiclassical theory, we should be able to reproduce general trends as, for example, the A-dependence of the pairing gap (Fig. 6.6). The fact that the Thomas–Fermi solution of the gap goes to zero at the classical turning point might, in this case, be a more severe shortcoming than in the case of the normal density, because of the pronounced surface character of the effect. Gradient terms (higher orders in $\hbar$) should therefore be included. However, no studies in this direction have been performed so far.

This concludes the semiclassical treatment of the static properties o nuclei, and we will turn next to the question of how we can use semiclassical considerations for the description of dynamic properties of the nucleus

13.3 The Dynamic Case

Until now we have treated static properties of the nucleus, like the nuclear density and the nuclear binding energy, by semiclassical methods. It is natural to extend these theories to the dynamic case such as RPA and time-dependent Hartree–Fock theory. We will therefore have to find the limit of such theories as $\hbar$ goes to zero. Several new features and problems arise in the dynamic case that are just at the initial stage of intensive study; thus we cannot be at all exhaustive nor can we give a rounded picture of the theory in this last section.

One of the basic equations in the dynamic case will be the classical limit of the time-dependent Hartree–Fock equations (12.9), which will lead us to nuclear fluid dynamic equations. A slightly different approach will be the derivation of such equations by a variational principle. In order to do that,

however, we need the functional $\tau[\rho]$ [see Eq. (13.57)] in the dynamic case. We will give an example of this, but this would appear rather difficult to find in general. Finally, we will discuss connections between the semiclassical results and those we obtained by sum rule considerations in Chapter 8.

13.3.1 The Boltzmann Equation

In order to take the classical limit of the TDHF equation (12.9), we have to take its Wigner transform, which we can do using the product rule (D.55). The result is [Gr 46, KK 76]

$$\frac{\partial f(\mathbf{q},\mathbf{p},t)}{\partial t}+\frac{2}{\hbar}f(\mathbf{q},\mathbf{p},t)\sin\left[\frac{\hbar}{2}\left(\overleftarrow{\nabla}_q\vec{\nabla}_p-\overleftarrow{\nabla}_p\vec{\nabla}_q\right)\right]H(\mathbf{q},\mathbf{p})=0, \quad (13.68)$$

with $f(\mathbf{q},\mathbf{p},t)$ being the Wigner transform of ρ [cf. (13.41)], and $H(\mathbf{q},\mathbf{p})$ the Wigner transform of the Hartree–Fock Hamiltonian. For simplicity, we will assume that the potential is local, but all formulae below can easily be generalized to nonlocal potentials, for example, of the Skyrme type (5.99), using Eq. (13.36). From (13.68) we easily get the classical limit:

$$\left[\frac{\partial}{\partial t}+\frac{\mathbf{p}}{m}\nabla_q-\frac{\partial V}{\partial \mathbf{q}}\nabla_p\right]f^{(0)}(\mathbf{q},\mathbf{p},t)=0. \quad (13.69)$$

This equation is most commonly known under the name *The Collisionless Boltzmann Equation* [UF 74, Hu 63] and $(1/(2\pi\hbar)^3)\,f^{(0)}(\mathbf{q},\mathbf{p},t)$ is therefore the classical distribution function. The physical content of (13.69) is nothing but the balance of what enters and leaves a classical phase space cell: the first term is the explicit change in time of the classical distribution function, the second one gives the change due to the velocity of the particles in the cell, and the third one represents the change of their momenta due to acceleration by a force. The Boltzmann equation is therefore nothing but the fact that the total time derivation of the classical distribution function which measures the number of particles in a given volume element of the phase space moving with time, must be zero: $df^{(0)}/dt=0$. This, of course, is only true if we have a gas of independent particles interacting with the walls of the mean field $V[\rho]$ but having no genuine two-body collisions. This is evidently inherent to TDHF theory, from which we started. If the particles interact, we have to consider that particles can be scattered into and out of the cell, leading to the so-called Boltzmann equation with collision term: $df^{(0)}/dt=I(f^{(0)})$, where $I(f^{(0)})$ is a nonlinear integral operator in $f^{(0)}$. This collision term can be obtained again in the limit $\hbar\to 0$ from an analogous quantum mechanical term, which is contained in the two-body correlation function (12.6) we neglected in deriving the TDHF equation [cf. Eq. (12.7) and the discussion thereafter]. How this is done in detail is explained in the book *Quantum Statistical Mechanics* by Kadanoff and Baym [KB 62, Sch 72a]. We will not be more explicit here since, as we have already discussed in the context

of TDHF theory, we will only be interested in slow—that is, energetically weakly excited—phenomena. Here we should, however, be able to neglect the collision term in a first approximation [KK 68], since as we discussed in Chapter 4, the mean free path for nucleons in a ground state nucleus is of the order of the nuclear dimensions.

Note that for the time-independent case, taking the classical limit of the TDHF equation consistent with the Thomas–Fermi approximation to the distribution function, $f^{(0)}(\mathbf{q},\mathbf{p})=\Theta(\lambda-\mathbf{p}^2/2m-V(\mathbf{q}))$ [cf. Eq. (13.40)], since $f^{(0)}$ is solution to (13.69) for static $V(\mathbf{q})$.

It is straightforward to write down the classical analogue to the RPA equation starting from (13.69), since we know (see Sec. 8.5) that we get the RPA by linearizing the TDHF equation. Taking the simplest ansatz for $V[\rho]=a\rho(\mathbf{q})+b\rho^2(\mathbf{q})$ [see Eq. (5.99)], we get after linearization of Eq. (13.69) with

$$\begin{aligned} f(\mathbf{q},\mathbf{p},t)&=f^{(0)}(\mathbf{q},\mathbf{p})+\delta f(\mathbf{q},\mathbf{p},t),\\ \rho(\mathbf{q},t)&=\rho^{(0)}(\mathbf{q})+\delta\rho(\mathbf{q},t),\\ \delta\rho(\mathbf{q},t)&=\frac{1}{(2\pi\hbar)^3}\int d^3p\,\delta f(\mathbf{q},\mathbf{p},t) \end{aligned} \tag{13.70}$$

the following equations.

$$\frac{\partial\delta f}{\partial t}+\frac{\mathbf{p}}{m}\frac{\partial\delta f}{\partial\mathbf{q}}-\frac{\partial V[\rho^{(0)}]}{\partial\mathbf{q}}\frac{\partial\delta f}{\partial\mathbf{p}}-\frac{\partial f^{(0)}}{\partial\mathbf{p}}\frac{\partial}{\partial\mathbf{q}}\left[(a+2b\rho^{(0)})\delta\rho\right]=0. \tag{13.71}$$

In infinite nuclear matter the third term of (13.71) vanishes and Eq. (13.71) has a special solution which can be given analytically [KB 62, No 64a]. Although this equation corresponds to the linearized Boltzmann equation, which is, in principle, a classical equation ($\hbar=0$), we can still keep the quantum mechanics correct with respect to the statistics. Thus in Eq. (13.71) for a Fermi (Bose) system, $f^{(0)}$ will be a Fermi (Bose) distribution function, and for a classical system a Boltzmann factor which completely changes the character of this equation. The analytical solution of (13.71) in the infinite matter case yields a $\delta f(\mathbf{q},\mathbf{p},t)$, showing that it adds to the isotropic static momentum distribution a non-isotropic distribution that is roughly egg-shaped and oscillating about its spherically symmetric equilibrium shape. This kind of vibration is called *zero sound*, in contrast to *ordinary sound* where the fermi sphere is just an oscillating translation and oscillating expansion, but its shape remains spherical (see, e.g., [KB 62, Chap. 7.4]).

In general, Eq. (13.71) cannot be solved analytically even for the infinite matter case, and for a finite system it is a rather complicated partial differential equation in seven variables which has not yet been solved for problems in nuclear physics.

13.3.2 Fluid Dynamic Equations from the Boltzmann Equation

In order to reduce the complexity of equations (13.69) and (13.71), one tries to transform them into coupled equations for the moments of $f(\mathbf{q}, \mathbf{p}, t)$ with respect to $\mathbf{p}$ in multiplying (13.69) and (13.71) by different powers of $\mathbf{p}$ and integrating over $\mathbf{p}$. This yields an infinite system of coupled partial differential equations in only four variables which are equivalent to the original equation. The simplification will consist of an appropriate decoupling of the hierarchy of coupled equations. This is a program which leads to the equations of *ordinary hydrodynamics* [UF 74, Hu 63], and we will see how in principle we have to modify the method in order to arrive at fluid dynamic equations which describe the propagation of *zero sound*.

For this purpose, we integrate Eq. (13.69) over $\mathbf{p}$, then multiply by p_i $(i=1,2,3)$ and by $\mathbf{p}^2$ and integrate again. After a few transformations the resulting five equations can be written in the form:

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial \mathbf{q}}(\rho \mathbf{u}) = 0, \tag{13.72}$$

$$m\rho \frac{Du_i}{Dt} \equiv m\rho\left(\frac{\partial u_i}{\partial t} + \mathbf{u}\frac{\partial u_i}{\partial \mathbf{q}}\right) = -\rho \frac{\partial V}{\partial q_i} - \sum_{j=1}^{3} \frac{\partial P_{ij}}{\partial q_j}. \tag{13.73}$$

$$\rho \frac{D}{Dt}\left(\frac{Q}{\rho}\right) + \frac{\partial}{\partial \mathbf{q}}\mathbf{w} = -\sum_{i,j=1}^{3} P_{ij} D_{ij}, \tag{13.74}$$

where we have introduced the following notation:

$u_i(\mathbf{q}, t) = \bar{p}_i / m$	the average velocity,
$U_i(\mathbf{q}, \mathbf{p}, t) = p_i/m - u_i$	deviation from the average velocity,
$Q(\mathbf{q}, t) = \frac{1}{2} m\rho \overline{U^2}$	the thermal energy density,
$P_{ij}(\mathbf{q}, t) = m\rho \overline{U_i U_j}$	the pressure tensor,
$w_i(\mathbf{q}, t) = \frac{1}{2} m\rho \overline{U_i \mathbf{U}^2}$	the heat current density, and
$D_{ij}(\mathbf{q}, t) = \frac{1}{2}(\partial u_i/\partial q_j + \partial u_j/\partial q_i)$	the rate of strain tensor,

where the average for an arbitrary quantity A is given by (neglecting the spin):

$$\bar{A}(\mathbf{q}, t) = \frac{\int d^3p\, A(\mathbf{p}, \mathbf{q}, t) \cdot f(\mathbf{p}, \mathbf{q}, t)}{\int d^3p\, f(\mathbf{q}, \mathbf{p}, t)} = \frac{1}{\rho} \frac{1}{(2\pi\hbar)^3} \int d^3p\, A \cdot f.$$

Equations (13.72–13.74) express nothing but the conservation laws for mass, momentum, and energy. However, these equations are, as they stand, useless, since they are not a closed system of equations. Only if we

could, for example, express the pressure tensor and the heat current density by the density ρ and velocity $\mathbf{u}$, respectively, would we get a closed system for unknown functions; this is exactly how we derive ordinary hydrodynamics. Here, however, we want to derive a fluid dynamical description of zero sound which is physically something completely different, as we have already briefly discussed (see also [Th 61b, KB 62, No 64a]).

In order to demonstrate the differences as clearly as possible, we first want to indicate very briefly what the physical argument is, so that we arrive at a closed system of equations coming from (13.72–13.74) for the case of ordinary hydrodynamics, and then discuss the different procedure we have to adopt to get the fluid dynamic description of zero sound and other low-energy collective nuclear phenomena.

(i) In ordinary hydrodynamics [UF 74, Hu 63] we assume that we are in a regime in which the characteristic time of typical collective motions we are interested in is very large compared to the time between successive collisions of the particles. As a consequence of these frequent collisions, the system will always be in *local equilibrium*. To understand the notion of local equilibrium, we have to realize that it was Boltzmann's achievement to show that his equation with a collision term has, for *any* initial distribution, a solution which approaches in the course of time the Maxwell–Boltzmann distribution [UF 74],

$$f_B^{(0)}(\mathbf{q}, \mathbf{p}) \propto e^{-(1/k_B T)[p^2/2m + V(\mathbf{q})]}. \tag{13.75}$$

This is a function with an isotropic momentum distribution [solution to the static part of (13.69)], which characterizes equilibrium. Accordingly, under local equilibrium we shall understand that in a given volume element, the velocity distribution of the particles is such as to reproduce the average velocity $\mathbf{u}(\mathbf{q}, t)$ of the element and to be of the Maxwellian form:

$$f_B^{(0)}(\mathbf{q}, \mathbf{p}, t) = \rho(\mathbf{q}, t)\left(\frac{2\pi\hbar^2}{mk_B T(\mathbf{q}, t)}\right)^{3/2} \exp\left[-(1/2mk_B T(\mathbf{q}, t))(\mathbf{p} - m\mathbf{u})^2\right]. \tag{13.76}$$

Here all quantities are only locally defined, that is, temperature $T(\mathbf{q}, t)$ and velocity $\mathbf{u}(\mathbf{q}, t)$ depend on position and time. With the assumption of local equilibrium (13.76), we can now decouple equations (13.72–13.74) in noting that

$$\begin{aligned} P_{ij} &= P\delta_{ij}, \qquad P = \rho k_B T, \\ w_i &= 0, \qquad Q = \tfrac{3}{2}P, \end{aligned} \tag{13.77}$$

whence it becomes obvious why we used the name pressure tensor. Equations (13.72–13.74) therefore reduce to five coupled equations for five unknown functions: these are the ideal or Euler hydrodynamical equations

[UF 74, Hu 63, LL 59, Vol. 6]*:

$$\frac{\partial \rho}{\partial t} + \operatorname{div}(\rho \mathbf{u}) = 0,$$

$$m\rho \frac{D}{Dt}\mathbf{u} = -\rho \frac{\partial V}{\partial \mathbf{q}} - \frac{\partial}{\partial \mathbf{q}} P, \tag{13.78}$$

$$\frac{D}{Dt}(\rho P^{-5/3}) = 0.$$

The assumption of local equilibrium and equations (13.78) constitute the first step of a systematic improvement (*Chapman Enskog development*), where we consider small deviations from local equilibrium: $f = f_B^0 \cdot (1+\eta)$; together with the Boltzmann equation with collision term this leads in the next step to the *Navier–Stokes equations* [UF 74].

(ii) In low-energy nuclear physics, the situation is quite different: the mean free path of the nucleons is of the order of the nuclear dimensions [KK 68] and we can therefore ignore the collisions to a first approximation (in a certain sense, a classical analogue would be a *Knudsen gas* [Kn 50]). This is the same assumption which led us to the time-dependent Hartree–Fock equation in the first place, that is, two-body correlations might be negligible to a good approximation. As a consequence, we can *not* use, at least for low-energy phenomena, the concept of local equilibrium to close the system of Eqs. (13.72–13.74). In principle, we would need the solution of (13.69) to be able to calculate the pressure tensor and the energy current density. Because of our inability to solve Eq. (13.69), however, we have to find some approximative solution. One obvious assumption is to imagine the change in time of the potential, that is, the density in (13.69), to be so slow that the potential locally in time can be considered a constant. Under the condition

$$\frac{\left|\frac{\partial V}{\partial t}\right|}{V(t)} \ll \omega, \tag{13.79}$$

where ω is a characteristic single-particle frequency, we can make the following ansatz for a solution of (13.69):

$$f^{(0)}(\mathbf{q}, \mathbf{p}, t) \simeq \Theta\left(\lambda - \frac{(\mathbf{p} - m\mathbf{u})^2}{2m} - V(\mathbf{q}, t)\right). \tag{13.80}$$

This, of course, is just the usual Thomas–Fermi result (13.40), where we however replaced the static potential $V(\mathbf{q})$ by the corresponding time-

* The introduction of a temperature and a Boltzmann distribution is actually not necessary for the derivation of Eqs. (13.78). It is sufficient to assume an isotropic velocity distribution, that is f depends only on $\mathbf{q}$, $(\mathbf{p} - m\mathbf{u})^2$, and t. We then have, in particular

$$P = \tfrac{2}{3}\left(\frac{n^*}{2m}\tau - \frac{m}{2}\rho u^2\right).$$

dependent one. With the ansatz (13.80), we are again able to close the system (13.72–13.74). For example, for the pressure tensor we obtain (including the spin)

$$P_{ij} = \frac{1}{5}\frac{\hbar^2}{m}(3\pi^2)^{2/3}[\rho(\mathbf{q}, t)]^{5/3}\delta_{ij} - \frac{m}{3}\rho u^2 \tag{13.81}$$

which yields a decoupling of Eqs. (13.72) and (13.73). We therefore want to describe some applications within this framework.

13.3.3 Application of Ordinary Fluid Dynamics to Nuclei

In Chapter 1 we studied in considerable detail the application of the *liquid drop* model to nuclei. The hydrodynamical equations were very much simplified by the assumptions of a homogeneous drop with a sharp surface, incompressibility, and irrotational flow. With (1.19), (1.30), and (1.34), we gave formulae for the excitation energies of multipolarity $\lambda \geqslant 2$. They did not agree very well with experiment and we will learn why below.

Steinwedel and Jensen [SJ 50] relaxed the assumption of incompressibility somewhat, supposing that only the total density is incompressible whereas proton and neutron densities individually are not. This idea was first used to describe the giant dipole (see Fig. 13.10) resonance, but can be generalized to polarization of all multipolarities ($T=1$ polarization modes); these modes have to be contrasted with the ones prescribed by the liquid drop model of Bohr, where also the individual densities are incompressible. For the giant dipole resonance this leads to the model proposed by Goldhaber and Teller [GT 48] (see Fig. 13.10), where the proton sphere vibrates against the neutron sphere. For the other multipolarities these are $T=1$ surface vibrations, where the proton surface deforms out of phase from the neutron surface. There also exist $T=0$ surface vibrations, where the whole nucleus deforms in a coherent way around its spherical equilibrium position. This is obviously only possible incompressibly for $L \neq 0$

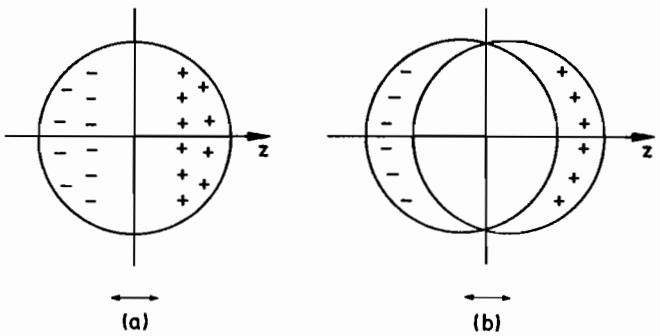


Figure 13.10. Dipole polarization of a spherical nucleus. Minus signs stand for proton defect, plus signs for neutron defect. (a) corresponds to the model of Steinwedel and Jensen, and (b) to the model of Goldhaber and Teller.

modes. For the $L=0$, $T=0$ breathing mode we have to give up the principle of incompressibility [EG 70].

From the above classification for the different possibilities of a liquid drop arises the question—for instance, for the $T=1$ modes—whether the Goldhaber–Teller model or the Steinwedel–Jensen model is physically realized. We want to present and contrast both models very briefly for the dipole case.

In the *Steinwedel–Jensen model* we assume that the total density has a sharp surface and stays constant in time:

$$\rho_0=\rho_p(\mathbf{r},t)+\rho_n(\mathbf{r},t), \tag{13.82}$$

where the proton density ρ_p and neutron density ρ_n vibrate around their equilibrium values $(Z/A)\rho_0$ and $(N/A)\rho_0$. We can therefore describe the process by the difference of the deviations of both densities:

$$\rho^{(1)}(\mathbf{r},t)=\delta\rho_p(\mathbf{r},t)-\delta\rho_n(\mathbf{r},t). \tag{13.83}$$

In the following, we assume that this is small compared to ρ_0 and derive equations in linear order of $\rho^{(1)}$. We can therefore linearize Eqs. (13.78), which yields for the equation of continuity,

$$\frac{\partial\rho^{(1)}}{\partial t}+\rho_0\nabla\mathbf{u}=0, \tag{13.84}$$

where $\mathbf{u}$ is the velocity field which corresponds to $\rho^{(1)}$; for the Euler equation we obtain

$$\frac{\partial\mathbf{u}}{\partial t}=-\frac{1}{m}\nabla\frac{\delta^2 e[\rho]}{\delta\rho^2}\rho^{(1)}, \tag{13.85}$$

where we used the fact that $\rho^{-1}m(\partial/\partial\mathbf{r})P=(\hbar^2/2m)(\partial/\partial\mathbf{r})(\delta/\delta\rho)\tau[\rho]$, taking for P_{ij} expression (13.81) and for $\tau[\rho]$ the Thomas–Fermi expression (13.32); that is, we have assumed an isotropic momentum distribution (13.76). Together with the force term in (13.78), we can combine this to the total energy density $e[\rho]=\tau+v$. The energy density can in this case be obtained from the Bethe–Weizsäcker formula (1.4); the only term therein depending on $\rho^{(1)}$ is the symmetry energy.

$$a_I\int_V\frac{(\rho_n-\rho_p)^2}{\rho_0}d^3r=a_I\frac{(N-Z)^2}{A}+\frac{a_I}{\rho_0}\int_V(\rho^{(1)})^2\,d^3r, \tag{13.86}$$

where we used $\int\rho^{(1)}\,d^3r=0$.

With (13.84), (13.85), and (13.86) we therefore get the following hydrodynamical wave equation for the dipole state.

$$\left(\Delta-\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\rho^{(1)}=0;\qquad c^2=\frac{2a_I}{m}. \tag{13.87}$$

We have to solve this equation with the proper spherical boundary condition, namely that there is no outgoing current at the nuclear surface. We therefore decompose $\rho^{(1)}(\mathbf{r},t)$ into its Fourier components $\rho^{(1)}(\mathbf{r},k)$ with

respect to time ($\omega = kc$). The angular dependence of $\rho^{(1)}$ is represented by spherical harmonics $Y_{\lambda\mu}$. We find that the lowest eigenvalues are realized for dipole vibrations ($\lambda = 1$). The corresponding radial equation has spherical Bessel functions as solutions. For the transition density we get

$$\rho^{(1)}(\mathbf{r}, k) \propto j_1(kr) Y_{1\mu}(\vartheta, \varphi). \tag{13.88}$$

The boundary condition that there is no radial current at the nuclear surface,

$$\left.\frac{\partial \rho^{(1)}}{\partial r}\right|_{r=R_0} \propto \left.\frac{\partial}{\partial r} j_1(kr)\right|_{r=R_0} = 0, \tag{13.89}$$

can only be satisfied for discrete k values. The lowest solution of this equation is

$$k_0 R_0 = 2.08,$$

yielding the frequency

$$\omega_0 = \frac{2.08c}{R_0}. \tag{13.90}$$

With $R_0 = r_0 A^{-1/3}$, (1.2), and (1.87), we obtain

$$\omega_0 = \frac{2.08}{r_0}\left(\frac{2a_I}{m}\right)^{1/2} A^{-1/3} \simeq 70\, A^{-1/3}[\mathrm{MeV}], \tag{13.91}$$

that is, the model gives roughly the right A dependence and almost the correct magnitude of ω_0 (see below).

The *Goldhaber–Teller model* can be best treated within the *Tassie model* [Ta 56a] for an inhomogeneous irrotational and incompressible fluid. The nucleus is supposed to vibrate around its spherical equilibrium position $\rho_0(r)$ and from (1.24–1.26), using the equation of continuity in linearized form, we get

$$\begin{aligned} \delta\dot{\rho} &= -\nabla(\rho_0(r)\mathbf{u}(\mathbf{r})) \\ &= -\frac{\partial \rho_0}{\partial r} u_r = -\frac{\partial \rho_0}{\partial r}\frac{\partial \Phi}{\partial r}. \end{aligned} \tag{13.92}$$

Taking the Fourier transform with respect to time, together with the expansion (1.27), we obtain the following expression for the transition density of multipole order λ.

$$\delta\rho_\lambda \propto \lambda r^{\lambda-1} Y_{\lambda 0} \frac{\partial \rho_0}{\partial r}, \qquad \lambda \neq 0. \tag{13.93}$$

The term $\lambda = 0$ is excluded because we obviously cannot treat the monopole vibration with the assumption of incompressibility. A similar expression (8.175), however, is obtained for the $\lambda = 0$ case in the *Werntz–Überall* model [WU 66].

For the $\lambda = 1$ (dipole) case, the Tassie model corresponds to the Goldhaber–Teller model because of the assumption of incompressibility.

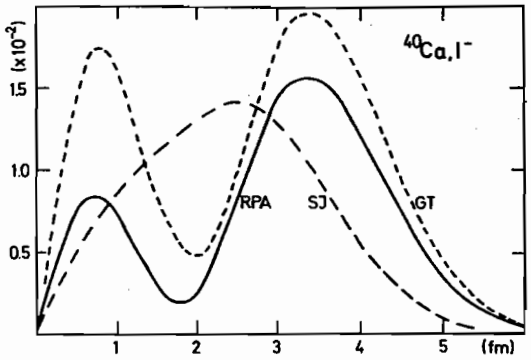


Figure 13.11. Transition densities for the dipole state of ^{40}Ca. The full curve stands for the transition density for the most collective dipole state of ^{40}Ca in the RPA. The dotted and dashed curves show those of the Goldhaber–Teller and Steinwedel–Jensen model, respectively. (From [SR 77].)

Then the only possible dipole motion is that shown in Fig. 13.10b. The A dependence of the excitation energy of the Goldhaber–Teller model is predicted [GT 48] to be $A^{-1/6}$. Experimentally we find that for light nuclei the dependence can be fitted by an $A^{-1/6}$ law, whereas for heavy nuclei it is a mixture of the Goldhaber–Teller and Steinwedel–Jensen types of motion. A more detailed investigation has been given by Myers et al. [MSK 77]. From sum rule considerations (Sec. 8.7.4), we know that the excitation operators for the Bohr–Tassie (Goldhaber–Teller) models are given by $F_\lambda = \sum r_i^\lambda Y_{\lambda 0}$, whereas in the Steinwedel–Jensen model they are given by $F_\lambda = \sum j_\lambda(kr_i) Y_{\lambda 0}$ [SR 77]. In Fig. 13.11 we compare the transition densities of the different models for the dipole state in ^{40}Ca with a microscopic RPA calculation. (For the relation of quantum mechanical densities with the $\rho^{(1)}$ defined here, compare Sec. 8.3.1.) We see that in this relatively light nucleus the vibration is mostly of the *GT* type. Similar comparisons in heavy nuclei [MSK 77] show that there the *SJ*-mode plays an increasingly important role.

From the above considerations, we see that ordinary hydrodynamics [isotropic momentum distribution (13.80, 13.81)] seems to work quite well for the explanation of the giant dipole resonances. We will see, however, in Section 13.3.6 that this, together with the breathing mode, is quite a special result, and that hydrodynamics have to be changed considerably in order to describe the giant resonances for all the other multipolarities.

The concept of isotropic momentum distribution has also been adopted by Holzwarth [Ho 77] and by Wong et al. [WMW 75, Wo 76, MWW 76, WWM 77, WM 77, WMW 77] to calculate the fluid dynamic counterpart of the TDHF slab collision discussed in Chapter 12. In Fig. 12.4 we show a comparison of both calculations for the final *kinetic* energy of relative motion of the outgoing slabs; zero kinetic energy means fusion. We see that below 7.6 MeV per projectile nucleon in the laboratory system, hydrodynamics always gives fusion, whereas TDHF gives resonant-like

scattering between 3.6 and 6.8 MeV and between 7.6 and 9 MeV. These "resonances" in the cross section are probably a typical quantum mechanical effect as discussed in Chapter 12, and can therefore not be reproduced by a semiclassical calculation. On the other hand, the semiclassical and quantum mechanical calculations are quite close to one another outside the resonance regions, as much in the final kinetic energies as in their density profiles. This, however, is essentially a one-dimensional calculation which cannot test what seems to be the main assumption of the ansatz (13.80), namely to give an *isotropic* momentum distribution.

We have discussed that the solution of (13.71) in the infinite matter case [KB 62, No 64a] gives an *anisotropic* momentum distribution. It seems difficult to obtain even approximately an analytic solution of (13.71) in the finite matter case, where first- and probably higher-order time derivatives of the potential are important. However, this we would need in order to close the system of equations (13.72–13.74).

Because of these difficulties, ordinary hydrodynamics has also been applied to more realistic situations. In any case, it is an interesting question whether for higher energies—for instance, in a heavy ion collision—the assumption of an isotropic momentum distribution, that is, local equilibrium, is not again a good approximation. Maruhn [Ma 77c] has performed a three-dimensional hydrodynamical calculation for the scattering of ^{16}O on ^{16}O with 18.8 MeV per nucleon in the laboratory system. He has also made a study of the Wigner distribution function of a full TDHF calculation for colliding slabs, which showed that at lower energies local equilibrium is not established. In view of this, and of what we have said about anisotropic momentum distribution above, the interpretation of low-energy ordinary hydrodynamics results should be done with great care.

Before we discuss more about this, let us first give a somewhat different derivation of fluid dynamical equations, which has some interesting aspects.

13.3.4 Variational Derivation of Fluid Dynamics

As we discussed in Chapter 12, quantum mechanics can be derived from a time-dependent variation principle [KK 76],

$$\delta I = 0,$$

where I is the action integral over the following time dependent Lagrangian.

$$I = \int_{t_1}^{t_2} dt\, \mathcal{L}[\Psi, \Psi^*]; \qquad \mathcal{L} = \langle \Psi | i\hbar \frac{\partial}{\partial t} - H | \Psi \rangle.$$

We will now make a special variational ansatz for $|\Psi\rangle$, which will lead us to fluid dynamic equations. The exact wave function can always be decomposed into real and imaginary parts in the following way (Φ_0, S real).

$$\Psi(1 \ldots A, t) = \Phi_0(1 \ldots A, t) e^{(im/\hbar) S(1 \ldots A, t)}. \tag{13.94}$$

Let us suppose that the phase is of the single-particle form

$$S(\mathbf{r}_1,\ldots,\mathbf{r}_A,t)=\sum_{i=1}^{A}s(\mathbf{r}_i,t)=\int s(\mathbf{r},t)\hat{\rho}\,d^3r, \tag{13.95}$$

with (D.1),

$$\hat{\rho}=\sum_{i=1}^{A}\delta(\mathbf{r}_i-\mathbf{r}),$$

which means, for instance, that if $|\Psi\rangle$ is a Slater determinant, the phases of all single-particle states are the same:

$$\rho(\mathbf{r},\mathbf{r}',t)=e^{i(m/\hbar)s(\mathbf{r},t)}\rho_0(\mathbf{r},\mathbf{r}',t)e^{-i(m/\hbar)s(\mathbf{r}',t)}, \tag{13.96}$$

$$\rho_0(\mathbf{r},\mathbf{r}',t)=\sum_{i=1}^{A}|\varphi_i(\mathbf{r},t)|\,|\varphi_i(\mathbf{r}',t)|.$$

Comparing (13.96) with Eq. (12.63), we see that the assumption (13.95) is equivalent to (Stringari [St 79] also treated nonlocal fields)

$$\langle\mathbf{r}|\chi|\mathbf{r}'\rangle=ms(\mathbf{r},t)\delta(\mathbf{r}-\mathbf{r}'), \tag{13.97}$$

that is, (13.96) is justified if we achieve a decomposition of the density matrix (12.82) in such a way that χ is approximately local [we recall in this context that the decomposition (12.63) is not unique].

With (13.95) we obtain for the action integral:

$$\begin{aligned}I&=\int_{t_1}^{t_2}dt\{\langle\Phi_0|-m\dot{s}|\Phi_0\rangle+i\hbar\langle\Phi_0|\dot{\Phi}_0\rangle-\langle\Psi|H|\Psi\rangle\}\\&=\int_{t_1}^{t_2}dt\Big\{-m\int d^3r\,\dot{s}\,(\mathbf{r},t)\rho_0(\mathbf{r},t)-\langle\Psi|H|\Psi\rangle\Big\},\end{aligned} \tag{13.98}$$

where because of (13.97) only the local ρ_0 enters. The term $\langle\Phi_0|\dot{\Phi}_0\rangle$ in (13.98) vanishes because $|\Phi_0\rangle$ corresponding to ρ_0 is time even. With (13.95), the expectation value of the Hamiltonian assumes the form:

$$\mathcal{H}=\langle\Psi|H|\Psi\rangle=\frac{m}{2}\int d^3r\,\rho_0(\mathbf{r},t)\mathbf{u}^2(\mathbf{r},t)+E_{\text{intr}}, \tag{13.99}$$

with

$$\mathbf{u}=\nabla s\quad\text{and}\quad E_{\text{intr}}=\langle\Phi_0|H|\Phi_0\rangle=\int d^3r e[\rho_0]. \tag{13.100}$$

In (13.99) we assumed the potential energy of H to be velocity independent (expression (13.99), however, is also correct for the Skyrme force in spite of its velocity dependence; see [GVV 76]). Integrating the first term of (13.98) by parts, we obtain

$$I=\int_{t_1}^{t_2}dt\Big\{m\int d^3r\,s\dot{\rho}_0-\int d^3r\Big(\frac{m}{2}\rho_0\mathbf{u}^2+e[\rho_0]\Big)\Big\}+C, \tag{13.101}$$

where I has taken on the form of a classical action integral [Go 59], with ρ_0 playing the role of the coordinate conjugate to the momentum. In (13.101)

we supposed that E_{intr} exists as a functional of the local density $\rho_0(r)$ [HK 64, MT 77].

Hamilton's equations, according to (13.101), are then

$$m\dot{\rho}_0 = \frac{\delta}{\delta s}\mathcal{H}, \tag{13.102a}$$

$$-m\dot{s} = \frac{\delta}{\delta \rho_0}\mathcal{H}. \tag{13.102b}$$

The first equation gives the equation of continuity ($\delta(\nabla s)^2 = 2\nabla s \delta \nabla s$; partial integration):

$$\dot{\rho}_0 = -\nabla(\rho_0 \nabla s) = -\nabla(\rho_0 \mathbf{u}), \tag{13.103}$$

and the gradient of the second equation yields a Euler-like equation (13.78)

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u}\nabla)\mathbf{u} = -\frac{1}{m}\nabla \frac{\delta e[\rho_0]}{\delta \rho_0}. \tag{13.104}$$

Since $\mathbf{u}$ is a gradient of the potentials, the flow associated with (13.96) and (13.97) is irrotational, a direct consequence of our assumption (13.95). We consequently should only accept solutions of (13.104) with $\nabla \times \mathbf{u} = 0$. The relation between variational formulation of fluid dynamics [GVV 76] and the one obtained from the Boltzmann equation is not obvious though certainly intimate. In any case, there remains the problem of how to close the system of equations—that is, in (13.73)—to find an appropriate $f^{(0)}$ to calculate the pressure tensor, and in (13.104) to construct the functional $e[\rho_0]$ for the dynamic case (we will see that knowing $f^{(0)}(t)$ also enables us to find the functional, so both problems are closely related).

13.3.5 Momentum Distribution of the Density ρ_0

In the preceding sections we conjectured several times that the assumption of a spherically symmetric momentum distribution for the fluid dynamic description of collective motion may not be valid for all cases. In Section 13.3.4 we saw that this concerns the calculation of the intrinsic energy E_{intr} (13.100) of the system, which enters as the potential energy in the fluid dynamic Hamiltonian of Eq. (13.99) very much in the same way as the constrained ground state enters as a potential of the Hamiltonian of the ATDHF theory (12.80). We consider, therefore, the density ρ_0 with which we have to calculate E_{intr} from the time-even part of the dynamic density $\rho(t)$. As in the ATDHF case, therefore, we should be able to calculate ρ_0 from a static *time-even** single-particle potential, and for the moment, in order to fix our ideas, we wish to think of a quadrupole constraint.

* By static we mean here that the time enters only in a trivial way in the sense that the constraint is different at different times, but at each moment we can solve a time-independent (static) Schrödinger equation for the calculation of ρ_0 [it need not necessarily correspond to the ground state density in this potential (cf. Chap. 12)].

Suppose we start from the ground state of A nucleons in a spherical (harmonic oscillator*) potential and start to deform it. As long as the deformations are so small (see the Nilsson diagram—Fig. 2.21) that there are *no level crossings* and all particles remain in their original orbits. However, because the wave functions are squeezed in one direction the momenta of the nucleons in this direction will be higher than along the long axis of the deformation. Therefore, if we stay in the same levels for all deformations and calculate the corresponding momentum distribution we will find that it depends strongly on the deformation. However, if we want the system to stay in its ground state, at each level crossing we must redistribute the particles in such a way that only the lowest levels are occupied. The reason the levels come down in the deformed potential is the fact that the corresponding wave functions are better adapted to the actual shape of the potential. However, this means that the wave functions, after redistributing the particles among the lowest levels, are less squeezed, consequently the momentum distribution becomes almost spherical again. Since the level crossings are quite frequent, we understand that the momentum distribution stays quite isotropic over the whole range of deformations *if* the system stays in the ground state of the deformed potential.

This fact can be analytically checked using the harmonic oscillator or semiclassically by using the three-dimensional locally harmonic form of the propagator (13.66) in Wigner space, for example, for a deformed Woods–Saxon potential, and calculating the corresponding momentum distribution using (13.11) and the saddle point method (13.2.4). We will find that, even for quite large deformations, the momentum distribution stays almost spherical, and that in any case it is a small effect.

The essential question now is whether in a (adiabatic) dynamic process the particles always stay in the ground state of the Hamiltonian $h(\rho_0)$ to at least a good approximation. This question is of course very much related to the adiabaticity of the process, and in ATDHF-theory (Sec. 12.3.1) we have seen that in the large amplitude case $[h(\rho_0), \rho_0] \propto \chi^2$—that is, if we start at $t=0$ with the system in its ground state $\rho_0(t=0)$—the system will in the course of time stay almost in the ground state if χ (the momentum) is small. We may therefore also say that the deviation from sphericity of the momentum distribution is a measure of the non-adiabaticity of the process. Whether adiabaticity is fulfilled for even the slowest imaginable process, spontaneous fission, is presently an open question. On the other hand, it is clear that the faster the process, the more particles will stay in the up-going (nonadapted) levels, as a quick glance at the Landau–Zener formula (12.149) shows. Therefore, the faster the process the greater the deformation of the momentum distribution.† In this sense we say that a deformed

* We like to think of a harmonic oscillator, because for this potential all the statements we are going to make may be verified analytically.

† Only for very high energies of the system, where the mean free path of the nucleons becomes much smaller than the nuclear dimensions, may the momentum distribution again equilibrate due to the frequent two-body collisions.

momentum distribution corresponding to ρ_0 is a dynamic effect. For which processes this is an important effect is presently the subject of intensive study.

There is, however, one kind of collective motion where the deformation of the momentum sphere is known [SH 78, HE 78] to play an essential role: the giant resonances, because their deformations are so small that there will be no level crossings. Since a deformed momentum distribution yields a higher kinetic energy than a spherical one, this effect will influence the spring constant of these harmonic vibrations (Chap. 8). In the deformation of the momentum sphere all particles are participating, therefore this is a volume effect leading to the characteristic $A^{-1/3}$-dependence of, for instance, the giant quadrupole resonance, which is in contrast to the original estimate of the liquid drop model (Sec. 1.4).

A deformed momentum sphere can, of course, also be simulated semiclassically in taking, for example, instead of the Thomas–Fermi expression (13.80) one which has *locally*—that is, at a certain place q in the nucleus—a deformed momentum distribution:

$$f^0 = \Theta\Big(\lambda(\mathbf{q}) - \frac{1}{2m}\big(ap_x^2 + bp_y^2 + cp_z^2\big)\Big); \qquad \lambda(\mathbf{q}) = \lambda - V(\mathbf{q}).$$

How such an expression can be derived mathematically will be shown in the following.

The time-even part ρ_0 of the density matrix (see above and Chap. 12) can always be expressed by the equilibrium density matrix $\bar{\rho}$ in the following form (12.66).

$$\rho_0(t) = e^{-(i/\hbar)\varphi(t)}\,\bar{\rho}\,e^{(i/\hbar)\varphi(t)}, \tag{13.105}$$

where φ is a Hermitian and time-odd operator. As an explicit example, we can consider a ρ_0 corresponding to a time-even Slater determinant; then (13.105) is simply a consequence of the Thouless theorem (E.26). The Wigner transform $\varphi = \varphi(\mathbf{q}, \mathbf{p}, t)$ is real and (cf. Sec. D.4) odd in $\mathbf{p}$. If we suppose, as we did in the case of χ (13.97), that the nonlocality of φ is weak, then we can expand in powers of $\mathbf{p}$ and stop after the first term:

$$\varphi(\mathbf{q}, \mathbf{p}, t) \simeq \mathbf{Q}(\mathbf{q}, t)\mathbf{p} + \cdots. \tag{13.106}$$

With this approximation, we get for the equation of continuity:

$$\begin{aligned}\dot{\rho}_0(\mathbf{q}, t) &= \frac{-i/\hbar}{(2\pi\hbar)^3}\int d^3p\,[\dot{\varphi}, \rho_0]_{\mathrm{W}} \\ &= -\rho_0(\mathbf{q}, t)\frac{\partial \dot{\mathbf{Q}}}{\partial \mathbf{q}} - \dot{\mathbf{Q}}\frac{\partial \rho_0(\mathbf{q}, t)}{\partial \mathbf{q}} + O(\hbar^2),\end{aligned} \tag{13.107}$$

where we used the expression (D.59) for the Wigner transform of a commutator. Comparison with (13.103) shows that with (13.106) we have, to lowest order, the following relation between $\mathbf{Q}$ and s.

$$\dot{\mathbf{Q}}(\mathbf{q}, t) = \nabla s(\mathbf{q}, t) = \mathbf{u}. \tag{13.108}$$

The Wigner transform of (13.105) is given by the product rule (13.36) as:

$$f_0(\mathbf{q},\mathbf{p},t)=\left[(e^{-i\varphi/\hbar})_{\mathrm{W}}e^{i\hbar\overleftrightarrow{\Lambda}/\hbar}\bar{f}(\mathbf{q},\mathbf{p})\right]e^{i\hbar\overleftrightarrow{\Lambda}/\hbar}(e^{i\varphi/\hbar})_{\mathrm{W}}. \tag{13.109}$$

The evaluation of this expression with (13.106) is somewhat facilitated if we go over to the **r** representation:

$$\rho_0(\mathbf{r},\mathbf{r}',t)=\exp\left[-\tfrac{1}{2}(\nabla_r\mathbf{Q}(\mathbf{r})+\mathbf{Q}(\mathbf{r})\nabla_r+\nabla_{r'}\mathbf{Q}(\mathbf{r}')+\mathbf{Q}(\mathbf{r}')\nabla_{r'})\right]\bar{\rho}(\mathbf{r},\mathbf{r}'), \tag{13.110}$$

where ∇_r acts on every function to its right. From (13.110) we see that the approximation (13.106), means for a position independent **Q** nothing but a scaling of the coordinates of the static density with a time-dependent velocity field $\mathbf{Q}:\mathbf{r}\to\mathbf{r}(t)=\mathbf{r}-\mathbf{Q}(t)$. The form (13.110) for ρ_0 is exactly what has been proposed by Holzwarth and Eckart [HE 78, 79]. Transforming (13.110) to relative and center of mass coordinates $(\mathbf{s},\mathbf{q})$, we obtain for the operator in the exponent up to second order in s:

$$\tfrac{1}{2}(\nabla\mathbf{Q}+\mathbf{Q}\nabla)_{\mathbf{q}+(\mathbf{s}/2)}+\tfrac{1}{2}(\nabla\mathbf{Q}+\mathbf{Q}\nabla)_{\mathbf{q}-(\mathbf{s}/2)}=\nabla_{q_i}Q_i(\mathbf{q})+s_kQ_{ik}(\mathbf{q})\nabla_{s_i}+\tfrac{1}{8}\nabla_{q_i}Q_{ijk}(\mathbf{q})s_js_k, \tag{13.111}$$

where we adopted the notation that we have to sum over indices, figuring twice ($i=1,2,3$), and

$$Q_{ijk\ldots}=\cdots\nabla_{q_k}\nabla_{q_j}Q_i(\mathbf{q}). \tag{13.112}$$

Using the expression (13.111) up to first order in **s** only, the density (13.110) can be brought into the general form (see footnote page 412):

$$\rho_0\left(\mathbf{q}+\frac{\mathbf{s}}{2},\mathbf{q}-\frac{\mathbf{s}}{2}\right)\equiv\rho_0(\mathbf{q},\mathbf{s})=\exp(\nabla_{q_i}Q_i)\exp(s_kA_{ik}\nabla_{s_i})\bar{\rho}(\mathbf{q},\mathbf{s}), \tag{13.113}$$

where the matrix A is given by:

$$A_{ik}=Q_{ik}(\mathbf{q},t)-\tfrac{1}{2}Q_jQ_{ikj}+\cdots. \tag{13.114}$$

The Wigner transform (13.33) of the density

$$f_0(\mathbf{q},\mathbf{p},t)=\int d^3s\,\rho_0(\mathbf{q},\mathbf{s})e^{-i\mathbf{p}\mathbf{s}/\hbar}$$

can now be calculated from (13.113) by partial integration to yield for (13.109):

$$\begin{aligned}f_0(\mathbf{q},\mathbf{p},t)&=\exp(\nabla_{q_i}Q_i)\exp(-A_{ii})\int d^3s\,\bar{\rho}(\mathbf{q},\mathbf{s})e^{s_kA_{ik}\nabla_{s_i}}e^{-i\mathbf{p}\mathbf{s}/\hbar}e^{-s_kA_{ik}\nabla_{s_i}}\cdot 1\\&=\exp(\nabla_{q_i}Q_i)\exp(-A_{ii})\int d^3s\,\bar{\rho}(\mathbf{q},\mathbf{s})\exp\left[\frac{i}{\hbar}\mathbf{p}e^{s_kA_{ik}\nabla_{s_i}}\mathbf{s}e^{-s_kA_{ik}\nabla_{s_i}}\right]\cdot 1\\&=\exp(\nabla_{q_i}Q_i)\exp(-A_{ii})\int d^3s\,\bar{\rho}(\mathbf{q},\mathbf{s})\exp\left[-\frac{i}{\hbar}p_k(e^{-A})_{kl}s_l\right]\\&=\exp(\nabla_{q_i}Q_i)\exp(-A_{ii})\,\bar{f}(\mathbf{q},\tilde{\mathbf{p}})\end{aligned} \tag{13.115}$$

where $f(\mathbf{q},\tilde{\mathbf{p}})$ is the Wigner transform of the density matrix $\bar{\rho}$ with the

momentum $\mathbf{p}$ replaced by

$$\tilde{p}_i = p_k (e^{-A})_{ki}. \tag{13.116}$$

From (13.115) and (13.116) we see that, to lowest order in the momentum expansion of φ (13.106), the Wigner transform of the time-even density ρ_0 can essentially be generally described by a deformation of the originally spherical momentum sphere of $\bar{\rho}$. This is exactly what we discussed at the beginning of this section. The treatment presented here is only valid for very small amplitude vibrations, because otherwise a scaling of the coordinates or the first-order expansion (13.106) is not valid. We thus have a semiclassical theory which should allow us to correctly calculate the features of the giant resonances. This will be shown in the next section.

In order to do this we first have to close the system of hydrodynamical equations. Using in (13.115) the Thomas–Fermi expression

$$\tilde{f}_{\mathrm{TF}} = \Theta\left(\lambda - \frac{\tilde{\mathbf{p}}^2}{2m} - \frac{\delta v[\rho_0]}{\delta \rho_0}\right), \tag{13.117}$$

we can calculate the pressure tensor (13.73), and together with (13.108) we obtain a closed system. For the case of the variational fluid dynamic equations (13.104), we have to calculate the functional $e[\rho_0]$, for instance, the functional $\tau[\rho_0]$. We will see that this is also possible using expression (13.115) and find that it will be different from the functional (13.57), which was derived under the assumption of a (almost) spherical momentum distribution.

13.3.6 Imposed Fluid Dynamic Motion

The velocity field $\mathbf{u}$ should, of course, be given by the solution of the fluid dynamic equations [EH 79, EHS 80]; in order to get an idea of what is going on, several authors [Be 74, 75, BS 76, SH 78, KTB 80, SBB 80, JL 80] studied the much less complicated problem of fluid motion, where one assumes certain types of $\mathbf{Q}$ fields; that is, the motion is imposed on the system [HE 78]. In Sec. 8.7 we have discussed that fields of the dipole or quadrupole type should come quite close to reality anyway because of some rule arguments. In order to keep things as simple as possible, we additionally assume that $\mathbf{Q}(\mathbf{r}, t)$ approximately factorizes in the case of small amplitude motion:

$$\mathbf{Q}(\mathbf{r}, t) = \alpha(t)\mathbf{v}(\mathbf{r}). \tag{13.118}$$

The velocity field $\mathbf{v}$ can now be chosen to be of the monopole, dipole, quadrupole, etc. form* (the fact that the dimensions are not the ones of a

*A further very interesting example of a velocity field is the so-called twist mode (cf. Chap. 8) with $\mathbf{v} = (yz, -xz, 0)$ [HE 77, 79]. It turns out that the local density stays unchanged, and only the momentum distribution becomes distorted (twisted).

velocity is only for convenience):

$$\begin{aligned} L=0: &\quad \mathbf{v}_0=\mathbf{r}=\nabla\frac{r^2}{2};\\ L=1: &\quad \mathbf{v}_1=(0,0,1)=\nabla z; \\ L=2: &\quad \mathbf{v}_2=(-x,-y,2z)=\tfrac{1}{2}\nabla(2z^2-x^2-y^2). \end{aligned} \tag{13.119}$$

As in (13.108) we see that the velocity fields can be expressed as the gradient of a potential and the motion will therefore be irrotational.

With (13.118) and (13.119) it is easy to calculate the distorted momenta (13.116) for the constrained motion:

$$\tilde{\mathbf{p}}_{L=0}=\begin{pmatrix} e^{-\alpha} & & 0\\ & e^{-\alpha} & \\ 0 & & e^{-\alpha}\end{pmatrix}\cdot\mathbf{p};\ \tilde{\mathbf{p}}_{L=1}=\mathbf{p};\ \tilde{\mathbf{p}}_{L=2}=\begin{pmatrix} e^{\alpha} & & 0\\ & e^{\alpha} & \\ 0 & & e^{-2\alpha}\end{pmatrix} \tag{13.120}$$

Thus we see that the momentum sphere stays spherical in the monopole case because of symmetry arguments; it also stays spherical in the dipole case because $\mathbf{v}_1$ does not depend on $\mathbf{r}$; in the quadrupole case, however, we get a highly non-isotropic momentum distribution. From (13.117) we obtain:

$$\tilde{p}^2<2m(\lambda-V)\equiv p_F^2(\mathbf{q},t). \tag{13.121}$$

Introducing polar coordinates for the momentum in the quadrupole case yields

$$p^2\left(e^{2\alpha}\sin^2\theta_p+e^{-4\alpha}\cos^2\theta_p\right)<p_F^2$$

or

$$p^2<\frac{p_F^2}{e^{2\alpha}\sin^2\theta_p+e^{-4\alpha}\cos^2\theta_p}\simeq p_F^2\left(1+\alpha(t)\sqrt{\frac{16\pi}{5}}\,Y_{20}(\theta_p)\right)^2, \tag{13.122}$$

that is, we obtain to lowest order in $\alpha(t)$ a time-dependent quadrupole deformation of the Fermi sphere. We should note, however, that this quadrupole deformation of the local momentum sphere is, to lowest order, characteristic for all kinds (apart from $L=0, 1$) of surface oscillations, and not just for the quadrupole vibration alone. This can be seen from the general expression (13.116) [HE 78] and understood by the general argument that ρ_0 must be even in $\mathbf{p}$, that is, the momentum distribution must be symmetric with respect to an inflection of $\mathbf{p}$ at the origin; the distribution is thus mainly of ellipsoidal shape. It should also be noted that the ellipse of the momenta is perpendicular to the one of the density for the $L=2$ case (see (13.135)). In a more heuristic way this can immediately be understood from the uncertainty principle.

The ansatz [(13.118) and (13.119)] imposes the form of the velocity field but leaves the time dependence $\alpha(t)$ open. We can use the fluid dynamic equations (13.102) for its determination. For this purpose we have to determine the functional $\tau[\rho_0]$ in the dynamic case; we present it here for the special case of the imposed fields (13.119), but the derivation can be easily generalized to arbitrary velocity fields. The kinetic energy density is given by (D.58):

$$\hbar^2\tau_0(\mathbf{q},t)=\frac{\hbar^2}{(2\pi\hbar)^3}\int d^3p\,\tau_0(\mathbf{q},\mathbf{p},t)\equiv\frac{2}{(2\pi\hbar)^3}\int d^3p\,p^2 f_0(\mathbf{q},\mathbf{p},t).$$

With the volume element in momentum space

$$d^3p = d^3\tilde{p}\,\det(\partial p_i/\partial \tilde{p}_k) = d^3\tilde{p}\,e^{\Delta u}, \tag{13.123}$$

and using (13.115) together with (13.116) and (13.119), we obtain

$$\tau_0 \equiv \tau_{0\alpha} = \frac{2}{(2\pi)^3}\frac{1}{\hbar^5} e^{\alpha v_L(\mathbf{q})\nabla_{\mathbf{q}}} \int d^3\tilde{p}\; p^2\, \bar{f}(\mathbf{q}, \tilde{\mathbf{p}}) e^{\Delta u}. \tag{13.124}$$

In order to calculate the integral in (13.124) we have to express p^2 by $\tilde{p}^2$, which can be achieved by inverting the relations (13.120):

$$p^2_{L=0} = e^{2\alpha}\tilde{p}^2, \qquad p^2_{L=1} = \tilde{p}^2, \tag{13.125}$$

$$p^2_{L=2} = \tilde{p}^2\left[\frac{2}{3}e^{-2\alpha} + \frac{1}{3}e^{4\alpha} + (e^{4\alpha} - e^{-2\alpha})\frac{1}{3}\sqrt{\frac{16\pi}{5}}\; Y_{20}(\theta_{\tilde{p}})\right].$$

With the Thomas–Fermi approximation (13.117) for $\bar{f}$ in (13.124), we get the dynamic kinetic energy density expressed in terms of the kinetic energy density $\bar{\tau}$ corresponding to the density matrix $\bar{\rho}$ (13.105)

$$\tau_{0\alpha}(\mathbf{r}, t) = e^{\Delta u} W_L(\alpha) e^{\alpha v_L \nabla_r} \bar{\tau}(\mathbf{r}), \tag{13.126}$$

with

$$W_{L=0} = e^{2\alpha}, \qquad W_{L=1} = 1, \qquad W_{L=2} = \tfrac{1}{3}(2e^{-2\alpha} + e^{4\alpha}). \tag{13.127}$$

Integrating (13.126) over $\mathbf{r}$ and integrating by parts, we obtain:

$$\int d^3r\, \tau_{0\alpha} = W_L(\alpha) \int d^3r\, \bar{\tau}(\mathbf{r}). \tag{13.128}$$

In the expressions (13.126) and (13.128), all the deformation dependence has been separated out and for $\bar{\tau}$ we can now use the functional derived under the assumption of a spherical momentum distribution (13.57). (Actually, we derived (13.126) only for the pure Thomas–Fermi case, but if we neglect the angle dependence of the potential energy we can show that (13.126) also holds in the case of the extended Thomas–Fermi theory [HE 78]).

From (13.108) and (13.118) we have

$$\nabla s = \dot{\alpha} \mathbf{v}_L \tag{13.129}$$

and the Hamiltonian (13.99) is therefore quadratic in the velocities $\dot{\alpha}$. Expanding E_{intr} (13.100) up to α^2 (small amplitude motion), Eq. (13.99) takes the form of a harmonic oscillator, and we can calculate the frequency Ω with which the nucleus vibrates if the motion (13.119) is imposed:

$$\Omega^2 = \frac{C}{B}, \tag{13.130}$$

with

$$B = m \int d^3r\, \bar{\rho}(r) \mathbf{v}_L^2(\mathbf{r}), \tag{13.131}$$

$$C = \frac{\partial^2}{\partial \alpha^2} E_{\text{intr}}\bigg|_{\alpha=0}. \tag{13.132}$$

We immediately see that the coefficient B corresponds to the one we derived in Chapter 1.4, Eq. (1.30), for irrotational flow. The restoring force

C has two contributions: one which comes from the kinetic energy ($E_{\text{intr}} = \int \tau + \int v$) and one from the potential energy, $C = C_\tau + C_v$. The part C_τ which comes from the kinetic energy can be calculated from (13.127) and (13.128):

$$C_\tau = \frac{\hbar^2}{2m} \frac{\partial^2}{\partial \alpha^2} \int d^3r \, \tau_{0\alpha}\Big|_{\alpha=0} = \frac{\hbar^2}{2m} \begin{cases} 4\int d^3r \, \bar{\tau} & L=0, \\ 0 & L=1, \\ 8\int d^3r \, \bar{\tau} & L=2. \end{cases} \tag{13.133}$$

In the case of the quadrupole vibration we obtain $B = m\int d^3r \, \bar{\rho}(\mathbf{r})r^2(2 - Y_{20}) = 2mA\langle r^2 \rangle$ and therefore, for the collective frequency,

$$\Omega^2_{L=2} = \frac{4\langle T \rangle}{mA\langle r^2 \rangle} + \frac{C_v}{mA\langle r^2 \rangle}. \tag{13.134}$$

The first term agrees with the sum rule result (8.166) and gives the correct A dependence, $\hbar\Omega_{L=2} \simeq 60A^{-1/3}$ [MeV] (8.167); to the second term are contributing only surface terms. This can be shown in the following manner. With (13.115) and (13.123) we have

$$\rho_0(\mathbf{q}, t) = \frac{2}{(2\pi\hbar)^3} \int d^3p \, f_0(\mathbf{q}, \mathbf{p}, t) = \exp(\nabla q_i Q_i) \frac{2}{(2\pi\hbar)^3} \int d^3\tilde{p} \, \tilde{f}(\mathbf{q}, \tilde{\mathbf{p}})$$

$$= \exp(\nabla_{q_i} Q_i)\bar{\rho}(\mathbf{q}). \tag{13.135}$$

For incompressible fluids ($\nabla \mathbf{v} = 0$) we obtain for arbitrary exponents σ

$$\rho^\sigma(\mathbf{r}, t) = \left(e^{\alpha \nabla \mathbf{v}} \bar{\rho}(\mathbf{r})\right)^\sigma = \bar{\rho}^\sigma(\mathbf{r} + \alpha \mathbf{v}) = e^{\alpha \nabla \mathbf{v}} \bar{\rho}^\sigma(\mathbf{r}), \tag{13.136}$$

and therefore:

$$\int d^3r \, \rho_0^\sigma(\mathbf{r}, t) = \int d^3r \, \bar{\rho}^\sigma(\mathbf{r}), \tag{13.137}$$

because from the exponential in (13.136) only the first term of its Taylor series contributes, the others being divergences of a vector field. The volume terms are thus α independent and do not contribute to the frequency. For the Skyrme force, only the terms of $v[\rho]$ containing derivatives of ρ—that is, surface terms— are different from zero in C_v; these yield a small $A^{-1/2}$ contribution to the collective energy, which is therefore negligible in first approximation compared to the kinetic energy terms of (13.134).

We now have the gratifying result that for the quadrupole mode, taking into account the deformation of the momentum sphere, yields the same result as that from the energy weighted sum, where we also impose the motion (8.166) to be of the quadrupole form. Imposing the form of the mode in both cases means that this state is an eigenstate of the system, and consequently the sum rule is automatically exhausted by this state because all others are orthogonal to it.

The reason we do not get the correct excitation energy in the pure liquid

drop model (1.34) can therefore be traced back to the fact that there we did not take into account the influence of the deforming walls of the mean field on the velocity distribution of the particles. The latter gives rise to the volume dependence (13.134) of the excitation energy also in the case of surface vibrations, and thus we obtain the correct A dependence.

On the other hand, we saw in Section 13.2.3 that the ordinary hydrodynamics of the liquid drop model work well for the dipole case. This is confirmed in the present model. Because of the coordinate independence of the velocity field (13.119) for $L=1$, no deformation of the momentum sphere occurs and the kinetic energy is the same in the static and dynamic case (13.128).

There is no contribution from the kinetic energy to the dipole excitation energy (13.133), and the fact that it nevertheless has a $A^{-1/3}$ dependence in the Steinwedel–Jensen model (Sec. 13.3.3) is only due to the fact that the restoring force comes from the symmetry energy (13.86). On the other hand, we should keep in mind that the dipole resonance is mostly of the Goldhaber–Teller type. The question of whether ordinary hydrodynamics is sufficient for the breathing mode can be decided by explicitly comparing the result we get with the functional $\tau[\rho]$ (13.57) to the one we obtained in (13.128). With (13.135), we obtain for this functional:

$$\tau_\alpha = C_1\left(e^{\alpha\nabla_r}\rho(\mathbf{r})\right)^{5/3} + C_2\frac{\left(\nabla e^{\alpha\nabla_r}\rho(\mathbf{r})\right)^2}{e^{\alpha\nabla_r}\rho(\mathbf{r})} + \cdots. \tag{13.138}$$

Integrating over $\mathbf{r}$ and integrating by parts gives the relation (13.128) for the first term in (13.138), so ordinary hydrodynamics can be used in the monopole case. We thus can conclude that $L=0, 1$ modes can be treated in the usual liquid drop model, but for all $L\geqq 2$ modes important modifications have to be included, due to the fact that a nucleus is a Fermi liquid rather than an ordinary fluid.

In a further step one should not, of course, impose the velocity field on the system, but rather this should come out as a solution of the fluid dynamic equations. A closed system of such equations can be obtained by either calculating the pressure tensor with (13.115) or in slightly generalizing the derivation of the functional $\tau[\rho]$ as it is given in Section 13.2.6, but without using the special ansatz (13.118) for the velocity field **Q** [HE 78]. It will be very interesting to see how close these solutions will come to the one we imposed on the system by (13.119) [HE 79].

Other interesting questions in this context are whether or not the approximation of irrotational flow is justified, whether the theory developed so far can be applied not only to small amplitude vibrations but also, for example, to heavy ion collisions. In this context it is important to establish the relation of this fluid dynamic description with ATDHF because of the many points in common.

13.3.7 An Illustrative Example

In order to investigate the validity of the assumptions that go into the derivation of nuclear fluid dynamics, it is very instructive to consider the following exactly soluble example. Suppose we have an initial nuclear density which corresponds to the ground state of A nucleons of a *deformed* harmonic oscillator. We let this density start to move at $t=0$ in the corresponding *spherical* harmonic oscillator which is given by the condition $\omega_0^3=\omega_x\omega_y\omega_z$. Since the wave functions of the deformed oscillator are not, of course, eigenfunctions of the spherical one, the solution of the corresponding Schrödinger equation will be time-dependent wave functions whose solutions are given by (10.89). Since the initial density as well as the time-dependent one will always be centered at the origin, the time dependence will only be a time-dependent scaling of the coordinates, that is, essentially a breathing of the width of the individual wave functions. Using the wave functions (10.89) we can write for the time dependent density [BS 79, SB 79]

$$\rho(\mathbf{r},\mathbf{r}',t)=e^{(i/\hbar)\chi(\mathbf{r},t)}\rho_0(\mathbf{r},\mathbf{r}',t)e^{-(i/\hbar)\chi(\mathbf{r}',t)}, \tag{13.139}$$

with

$$\chi(\mathbf{r},t)=\frac{m}{2}\frac{d}{dt}\left(\ln\xi_x x^2+\ln\xi_y y^2+\ln\xi_z z^2\right) \tag{13.140}$$

and

$$\xi_j^2=(\sigma_j+\tau_j\cos\omega_0 t)\qquad (j=x,y,z),$$

where σ and τ are defined in (10.89c). The density ρ_0 is of course the *time-even* part of ρ. Since ρ_0 can be written as ($\bar{b}_j$ are measured in units of $b_0=\sqrt{\hbar/m\omega_0}$)

$$\rho_0(\mathbf{r},\mathbf{r}',t)=\rho_0\left(\frac{x}{\bar{b}_x(t)},\frac{y}{\bar{b}_y(t)},\frac{z}{\bar{b}_z(t)},\frac{x'}{\bar{b}_x(t)},\frac{y'}{\bar{b}_y(t)},\frac{z'}{\bar{b}_z(t)}\right), \tag{13.141}$$

we can express ρ_0 in the following way.

$$\rho_0(\mathbf{r},\mathbf{r}',t)=e^{\frac{1}{2}[Q(\mathbf{r},t)\overrightarrow{\nabla}_r+\overrightarrow{\nabla}_{r'}Q]}\rho_{\text{stat}}(\mathbf{r},\mathbf{r}')e^{\frac{1}{2}[Q\overleftarrow{\nabla}_r+\overleftarrow{\nabla}_{r'}Q]}, \tag{13.142}$$

where the arrows on the three-dimensional gradient operators indicate the direction in which they are acting and ρ_{stat} is the density matrix of the ground state of the A nucleons in the *spherical* harmonic oscillator. As we see from the relation $\exp(\alpha x\partial/\partial x)f(x)=f(e^{\alpha}x)$, the velocity field in this example is given by

$$Q_j=\alpha_j(t)\cdot v_j(\mathbf{r}),\qquad (j=x,y,z), \tag{13.143}$$

$$\alpha_j(t)=-\ln\bar{b}_j(t),\qquad \mathbf{v}(\mathbf{r})=\mathbf{r}. \tag{13.144}$$

From this example we see that the main assumptions for the derivation of fluid dynamic equations, namely that the operators χ (12.63) and φ (12.66)

are local and only slightly nonlocal, respectively, are exactly fulfilled in this model. Since, after all, a harmonic oscillator is not such a bad approximation, we should expect that also for more realistic cases, operators χ and φ can be found which are almost local. We should stress the point, however, that the χ as it is given in (13.140) is not equivalent to the definition of χ^{BV} of Baranger and Véneroni [BV 78]. On the contrary, χ^{BV} is rather nonlocal, even in this model. To give an estimate we calculate the Wigner transform of χ^{BV} to lowest order in $\hbar$ which should, as usual, give the right order of magnitude for the nonlocality [see Eq. (D.24)]. For small χ's and using (D.24) and the Hermiticity of X we obtain [SB 79]

$$\chi^{BV}(\mathbf{q},\mathbf{p}) = \frac{\hbar^2}{2}\rho(\mathbf{q},\mathbf{p})\overleftrightarrow{\Lambda}\rho(\mathbf{q},-\mathbf{p}). \tag{13.145}$$

This means that the range of the nonlocality of χ^{BV} is of the same order as that for the density, which is usually quite large [$\sim$2 fm]. We conclude that the definition of χ as given by Baranger and Véneroni is not optimal for a fluid dynamic treatment, but it is an open question as to how to define the most local χ in general.